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ODEWorld: A Continuous Predictive Architecture via Physical-Time Flow

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In the physical world we inhabit, space and time are fundamentally continuous. However, existing machine learning paradigms for world modeling are largely confined to discrete-time prediction and inference, thereby exhibiting significant inefficiency in capturing the underlying dynamics of the physical world. To bridge this gap, we introduce Physical-Time Flow (**PT-Flow**), a novel approach that learns a continuous latent velocity field operating in physical time. Crucially, the underlying dynamics of sequential data are parameterized by an ordinary differential equation (ODE) embedded in a well-structured dynamical representation space. Under this paradigm, the prediction of the future can be recast as temporal integration via an ODE solver in the compressed latent space. Building upon PT-Flow, we construct **ODEWorld**, a continuous-time latent world model that is both efficient and versatile. By extracting time-variant features and enforcing ODE properties on both the dynamical representation space and the latent velocity field, ODEWorld effectively addresses the long-standing representation collapse issue in the latent world model literature. This also enables high-quality image reconstruction with ODEWorld even after long-horizon prediction. Moreover, its continuous nature allows for arbitrary temporal resolution and even backward prediction, which is impossible for most discrete-time models. Lastly, benefiting from the learned dynamics-centric latent space, ODEWorld can provide rich planning-oriented information to facilitate downstream policy learning. Comprehensive experiments across multiple simulation and real-world datasets demonstrate that ODEWorld successfully reconciles planning-conducive dynamics abstraction with visual realism, excelling in both video generation and robotic control. More results can be seen at Project Website.

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1 Introduction

The physical world we inhabit is fundamentally continuous in both space and time. In our daily life, objects move, interact, and collide in continuous time, governed entirely by underlying physical laws; meanwhile, biological organisms, such as humans and animals, are naturally attuned to the perception, understanding, and decision-making of continuous transitions. Despite this, contemporary AI models almost exclusively rely on discrete-time modeling paradigms, such as modeling video as sequences of image frames (Ho et al., 2022; Bruce et al., 2024; Brooks et al., 2024; He et al., 2026; Wang et al., 2026b) and solving decision-making tasks in discrete time steps (Kim et al., 2026; Sutton et al., 1998; Hansen et al., 2022; Li et al., 2025; Ye et al., 2026). This inherent discrepancy makes modern predictive architectures highly inflexible when handling irregularly sampled data, imposing strict requirements on data for temporal alignment during both training and generation (Ha and Schmidhuber, 2018; Hafner et al., 2023). Furthermore, simply relying on learning discrete-time pixel/sensor measurement transitions ignores the fundamental continuity in physics-grounded systems, causing current models to function primarily as "video-mimicking", and become inefficient in internalizing the intrinsic, higher-order physics inherent in reality (Cellier and Greifeneder, 2013; Krishnapriyan et al., 2023).

To bridge this gap, we introduce the concept of Physical-Time Flow (PT-Flow). At its core, PT-Flow conceptualizes the underlying dynamics in the data as a continuous velocity field in a well-structured latent space, where the latent state transition is smoothly governed by an ordinary differential equation (ODE) along

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ODEWorld：基于物理时间流的连续预测架构

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我们生活的物理世界，其空间和时间本质上是连续的。然而，现有世界建模的机器学习范式大多局限于离散时间预测与推断，因而在捕捉物理世界的底层动力学时效率明显不足。为弥合这一差距，我们提出物理时间流（Physical-Time Flow，PT-Flow），学习一个在物理时间中运行的连续潜在速度场。关键在于，序列数据的底层动力学由嵌入结构良好的动力学表征空间中的常微分方程（ODE）参数化。在该范式下，未来预测可转化为在压缩潜空间中通过 ODE 求解器进行时间积分。基于 PT-Flow，我们构建了高效且多用途的连续时间潜在世界模型 ODEWorld。通过提取随时间变化的特征，并在动力学表征空间和潜在速度场上施加 ODE 性质约束，ODEWorld 有效解决了潜在世界模型文献中长期存在的表征坍塌问题。这也使它在长时域预测后仍能实现高质量图像重建。此外，连续性允许任意时间分辨率，甚至反向预测，而多数离散时间模型无法做到。最后，得益于学得的动力学中心潜空间，ODEWorld 可提供丰富的规划相关信息，促进下游策略学习。在多个模拟与真实世界数据集上的全面实验表明，ODEWorld 成功兼顾利于规划的动力学抽象和视觉逼真度，在视频生成与机器人控制上均表现出色。更多结果见项目网站。

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1 引言

我们生活的物理世界在空间和时间上本质连续。日常生活中，物体在连续时间中运动、交互和碰撞，完全受底层物理定律支配；人类和动物等生物也天然适应对连续变化的感知、理解与决策。尽管如此，当代人工智能模型几乎完全依赖离散时间建模，例如将视频建模为图像帧序列（Ho et al., 2022; Bruce et al., 2024; Brooks et al., 2024; He et al., 2026; Wang et al., 2026b），以及在离散时间步中解决决策任务（Kim et al., 2026; Sutton et al., 1998; Hansen et al., 2022; Li et al., 2025; Ye et al., 2026）。这种根本差异使现代预测架构处理不规则采样数据时缺乏灵活性，训练和生成均对时间对齐有严格要求（Ha and Schmidhuber, 2018; Hafner et al., 2023）。而且，仅学习离散时间的像素／传感器测量转移，忽略了物理系统的根本连续性，使模型主要成为“视频模仿器”，难以高效内化现实中固有的高阶物理规律（Cellier and Greifeneder, 2013; Krishnapriyan et al., 2023）。

为弥合这一差距，引入物理时间流（PT-Flow）。其核心是把数据底层动力学看作结构良好潜空间中的连续速度场，潜在状态转移通过常微分方程（ODE）平滑支配，并沿

actual physical time. To make such a continuous modeling framework possible, PT-Flow features two critical designs: 1) *dynamical representation decoupling* and 2) *direct first-order supervision*. The former decouples the time-varying, dynamics-related information from the data into a clean, ODE-compatible dynamical representation space, getting rid of the nuanced background and time-invariant details for efficient learning. The latter directly supervises the latent velocity field with the approximated ground truth latent velocities through an elegant Jacobian-vector product (JVP) projection. The above two designs lead to a new predictive architecture, in which the prediction of the future can be recast as temporal integration via an ODE solver in the compressed latent space, in some sense similar to flow matching (Lipman et al., 2023), but operating in physical time.

Building upon PT-Flow, we construct a continuous-time latent world model, called **ODEWorld**, which embodies many desirable features. First, it disentangles dynamical representation by conditioning the encoder and decoder directly on static context, thereby offloading the burden of dynamical representation learning, allowing the latent space to focus exclusively on capturing essential dynamical changes. Second, ODEWorld circumvents the long-standing challenge of representation collapse in Joint Embedding Predictive Architecture (JEPA)-based frameworks (Assran et al., 2023; Bardes et al., 2024; Assran et al., 2025), where the latent space can degenerate into trivial solutions due to the indirect supervision of the coupled encoder and latent predictor. While some recent efforts (Wang et al., 2026a; Maes et al., 2026) have added external regularization schemes to mitigate this issue, however, at the expense of sacrificing model expressiveness. By contrast, ODEWorld imposes direct supervision on the latent velocity field while harmoniously regularizing the ODE properties of the dynamical representations. These architectural choices empower ODEWorld to achieve high-fidelity image reconstruction with an extremely compact model, even across extended temporal horizons. Moreover, its continuous nature also enables arbitrary temporal resolution and even backward prediction–capabilities fundamentally absent in most discrete-time predictive models.

Empirical evaluations across diverse simulation and real-world benchmark datasets, including LIBERO (Liu et al., 2023) and Agibot-World (Bu et al., 2025), demonstrate that ODEWorld effectively reconciles planning-conducive dynamics abstraction with visual realism. We show that the learned latent velocity field produces interpretable and smooth transformation paths between the given initial and goal states. This well-behaved continuous latent flow enables temporally consistent, long-horizon predictions by leveraging off-the-shelf ODE solvers within a highly compact latent manifold. Moreover, prediction by solving the continuous-time ODE naturally supports temporal super-resolution and robust video generation from irregularly sampled or dropped-frame data, where discrete-time models inherently fail. Lastly, robotic control experiments on the LIBERO Benchmark also reveal that ODEWorld extracts rich, planning-centric representations that can greatly facilitate downstream policy learning. Ultimately, these capabilities establish ODEWorld as a highly versatile architecture, setting a new continuous recipe for both high-fidelity video synthesis and effective robotic policy learning.

2 Related Work

Predictive world models. Recent video generation models have achieved notable progress and shown great promise as scalable generative simulators of the physical world (Brooks et al., 2024; Yang et al., 2024; Yang et al.; Bjorck et al., 2025; Jang et al., 2025). These models learn to predict future observations as a surrogate for physical dynamics, producing full trajectory videos (Du et al., 2023; Zhou et al., 2024) or subgoal images (Black et al., 2024; Zhao et al., 2025). These models have also garnered substantial attention from the robotics community, which translates their video prediction capability for control. A first line of methods adopts a planning-and-control paradigm, where actions are inferred from predicted videos, via inverse dynamics models (Du et al., 2023; Zhou et al., 2024), goal-conditioned policies (Black et al., 2024; Li et al., 2026; Pai et al., 2025; Zheng et al., 2024; Li et al., 2024a), or direct pose extraction using off-the-shelf foundation models (Liang et al., 2025a; Chen et al., 2025). A second line of work instead integrates action generation into the world model, learning the joint distributions over future observations and actions (Li et al., 2025; Ye et al., 2026; Kim et al., 2026; Cheang et al., 2024; Bi et al., 2025; Guo et al., 2024; Zhu et al., 2025). However, all of these models suffer from architectural heaviness due to costly pixel-level reconstruction to derive world understanding, which have long been critiqued for their fundamental computational inefficiency.

To address the efficiency issue, another branch of work focuses on learning latent world models, where

实际物理时间。为使连续建模可行，PT-Flow 包含两项关键设计：1）动力学表征解耦；2）直接一阶监督。前者将数据中随时间变化、与动力学相关的信息解耦到干净且兼容 ODE 的动力学表征空间，去除细微背景和时间不变细节，以高效学习。后者通过巧妙的雅可比—向量乘积（JVP）投影，用近似真实潜在速度直接监督潜在速度场。由此形成新预测架构，将未来预测转化为在压缩潜空间中借助 ODE 求解器进行时间积分；在某种意义上类似流匹配（Lipman et al., 2023），但运行于物理时间。

基于 PT-Flow，我们构建了名为 ODEWorld 的连续时间潜在世界模型，它具有多项理想性质。首先，它通过让编码器和解码器直接以静态上下文为条件，解耦动力学表征，减轻表征学习的负担，使潜空间专注于捕捉关键动力学变化。其次，它绕开了基于联合嵌入预测架构（JEPA）的框架长期存在的表征坍塌挑战（Assran et al., 2023; Bardes et al., 2024; Assran et al., 2025）：编码器和潜在预测器相互耦合、仅受间接监督，可能使潜空间退化为平凡解。近期一些工作（Wang et al., 2026a; Maes et al., 2026）加入外部正则化缓解问题，却牺牲模型表达能力。ODEWorld 则直接监督潜在速度场，同时协调地正则化动力学表征的 ODE 性质。这些架构选择使极为紧凑的模型在较长时间范围内仍能实现高保真图像重建。此外，连续性支持任意时间分辨率，甚至反向预测，这些能力在多数离散时间预测模型中根本不存在。

在 LIBERO（Liu et al., 2023）和 Agibot-World（Bu et al., 2025）等多样的模拟与真实世界基准上的评估表明，ODEWorld 有效兼顾了有利于规划的动力学抽象与视觉逼真度。学得的潜在速度场在给定初始状态与目标状态间生成可解释、平滑的变化路径。在高度紧凑的潜在流形中使用现成 ODE 求解器，这种行为良好的连续潜在流支持时间一致的长时域预测。此外，通过求解连续时间 ODE，预测天然支持时间超分辨率，以及从不规则采样或丢帧数据鲁棒生成视频，而离散时间模型在此存在固有缺陷。LIBERO 上的机器人控制实验还表明，ODEWorld 提取丰富的规划中心表征，大幅促进下游策略学习。这些能力使 ODEWorld 成为高度通用的架构，为高保真视频合成和有效机器人策略学习提供新的连续时间方案。

2 相关工作

预测式世界模型。近期视频生成模型取得显著进展，显示出作为可扩展物理世界生成模拟器的巨大潜力（Brooks et al., 2024; Yang et al., 2024; Yang et al.; Bjorck et al., 2025; Jang et al., 2025）。这些模型通过预测未来观测来代理物理动力学，生成完整轨迹视频（Du et al., 2023; Zhou et al., 2024）或子目标图像（Black et al., 2024; Zhao et al., 2025）。机器人领域也广泛关注如何把视频预测能力用于控制。第一类方法采用规划与控制范式，通过逆动力学模型（Du et al., 2023; Zhou et al., 2024）、目标条件策略（Black et al., 2024; Li et al., 2026; Pai et al., 2025; Zheng et al., 2024; Li et al., 2024a），或用现成基础模型直接提取姿态（Liang et al., 2025a; Chen et al., 2025），从预测视频推断动作。第二类方法把动作生成整合到世界模型中，学习未来观测与动作的联合分布（Li et al., 2025; Ye et al., 2026; Kim et al., 2026; Cheang et al., 2024; Bi et al., 2025; Guo et al., 2024; Zhu et al., 2025）。然而，这些模型都通过代价高昂的像素级重建获取世界理解，造成架构沉重；其根本计算效率不足长期受到批评。

为解决效率问题，另一类工作聚焦学习潜在世界模型，其中

predictions are performed in a compact representation space (Assran et al., 2025; Maes et al., 2026; Hu et al., 2025; He et al., 2024; Zheng et al., 2025; Liang et al., 2025b; Wang et al., 2025; Liu et al., 2025; Kang et al., 2026). These approaches construct latent representations through various objectives, including image reconstruction (Hu et al., 2025; Xie et al., 2025), contrastive learning (Liu et al., 2025; Kang and Kuo, 2025; Nair et al., 2023), and explicit motion prediction (Wen et al., 2024; Zheng et al.; Gao et al., 2025), aiming to capture task-relevant structure while reducing the complexity of pixel-space modeling. An outstanding line of work is the JEPA family (Assran et al., 2023; Wang et al., 2026a; Maes et al., 2026; Assran et al., 2025; Balestriero and LeCun, 2025), which jointly learns to encode observations into a compact latent space and models transition dynamics by predicting future latents. While JEPA provides a compelling paradigm for latent world modeling, these methods can suffer from representation collapse, where the latent space degenerates toward trivial solutions, thus requiring carefully designed training strategies and regularization at the expense of expressivity loss (Wang et al., 2026a; Maes et al., 2026). Both the aforementioned video world models and latent world models are exclusively constructed in discrete-time, which potentially become inefficient when capturing the fundamental higher-order physics in reality (Cellier and Greifeneder, 2013; Krishnapriyan et al., 2023). In contrast, our proposed ODEWorld is directly modeled in continuous time while avoiding the representation collapse issue in JEPA frameworks by design, thereby offering great efficiency to model the physical world.

ODE dynamics modeling. Modeling continuous dynamical systems through ordinary differential equations has found extensive applications across diverse scientific domains, including physics (Anderson and Wendt, 1995; Grigorenko and Grigorenko, 2003; Brunton et al., 2016), climate science (Trenberth and Trenberth, 1992), and neuroscience (Izhikevich, 2007). One prominent methodology involves explicitly constructing physics-informed learnable ODE dynamics (Brunton et al., 2016; Champion et al., 2019; Greydanus et al., 2019). While these methods effectively embed known invariant properties into the network structure, they strictly require explicit prior knowledge of the true underlying governing equations or a given set of analytical terms to model the system dynamics, thereby severely limiting their expressiveness. Within this data-driven paradigm, classic approaches like Koopman operator theory (Mezić, 2005) are frequently employed in the control community. However, it also suffers from computationally expensive coordinate transformations and remains fundamentally restricted to an approximation of the true system dynamics. To circumvent the reliance on prior knowledge, neural ODE networks that learn neural ODE dynamic models directly from observation data have received increasing interest (Chen et al., 2018; Dupont et al., 2019; Liu et al., 2019; Huh et al., 2020). Despite being promising, training such neural ODE networks often relies on computationally expensive adjoint methods (Chen et al., 2018), which is hard to scale and often suffers from instability during training. By contrast, in our proposed PT-Flow paradigm, we directly supervise the latent ODE velocity field with the approximated ground truth latent velocities, which reconciles the preservation of ODE properties with efficient and scalable training.

3 Physical-Time Flow

To go beyond the discrete-time modeling to a new architecture capturing continuous-time physical reality, we introduce the concept of **Physical-Time Flow (PT-Flow)**. The core idea of PT-Flow is to capture the underlying dynamics of high-dimensional data as a continuous velocity field within a proper latent space. Let $s_t \in \mathbb{R}^{d_s}$ denote the state input at physical time t, and $z_t \in \mathbb{R}^{d_z}$ be a properly encoded latent representation of s_t. Here, t represents the physical time elapsed after starting at the initial state s_0. In our setting, s_0 can be flexibly selected from any frame in a demonstration rather than only the first frame. Accordingly, t is defined as the relative temporal offset between s_0 and s_t. To capture the time-evolving dynamics of the physical system, a straightforward approach is to learn a time-dependent velocity field defined via the ordinary differential equation:

$$\frac{dz_t}{dt} = v_\theta(z_t, t; z_0, c) \implies z_T = z_0 + \int_0^T v_\theta(z_t, t; z_0, c)dt \tag{1}$$

where z_0 is the latent representation of the initial state s_0 and c is the target conditioning (e.g., a goal state or a language instruction), which jointly specify the start and end state of the system. If we can learn such a latent velocity field v_θ, then future prediction can be redefined as a temporal integration task over

预测在紧凑表征空间中进行（Assran et al., 2025; Maes et al., 2026; Hu et al., 2025; He et al., 2024; Zheng et al., 2025; Liang et al., 2025b; Wang et al., 2025; Liu et al., 2025; Kang et al., 2026）。这些方法通过图像重建（Hu et al., 2025; Xie et al., 2025）、对比学习（Liu et al., 2025; Kang and Kuo, 2025; Nair et al., 2023）或显式运动预测（Wen et al., 2024; Zheng et al.; Gao et al., 2025）等目标构建潜在表征，旨在捕捉任务相关结构，同时降低像素空间建模复杂度。JEPA 系列是其中的重要分支（Assran et al., 2023; Wang et al., 2026a; Maes et al., 2026; Assran et al., 2025; Balestriero and LeCun, 2025），它联合学习将观测编码到紧凑潜空间，并通过预测未来潜变量建模转移动力学。尽管 JEPA 提供了有吸引力的潜在世界建模范式，却可能发生表征坍塌，使潜空间退化为平凡解，因此需要精心设计的训练策略与正则化，并损失表达能力（Wang et al., 2026a; Maes et al., 2026）。上述视频世界模型和潜在世界模型均完全基于离散时间构建，在捕捉现实中的基本高阶物理规律时可能效率不足（Cellier and Greifeneder, 2013; Krishnapriyan et al., 2023）。我们提出的 ODEWorld 则直接在连续时间建模，并通过设计避免 JEPA 框架的表征坍塌问题，从而高效建模物理世界。

ODE 动力学建模。用常微分方程建模连续动力系统，已广泛用于物理（Anderson and Wendt, 1995; Grigorenko and Grigorenko, 2003; Brunton et al., 2016）、气候科学（Trenberth and Trenberth, 1992）及神经科学（Izhikevich, 2007）等领域。一种重要方法是显式构建物理信息引导的可学习 ODE 动力学（Brunton et al., 2016; Champion et al., 2019; Greydanus et al., 2019）。这些方法能有效把已知不变性嵌入网络结构，但严格要求真实支配方程的显式先验，或给定一组解析项来建模系统动力学，严重限制了表达能力。在数据驱动范式中，Koopman 算子理论（Mezić, 2005）等经典方法常用于控制领域，但同样需要昂贵的坐标变换，且本质上只能近似真实系统动力学。为摆脱先验依赖，直接从观测数据学习神经 ODE 动力学模型的网络日益受到关注（Chen et al., 2018; Dupont et al., 2019; Liu et al., 2019; Huh et al., 2020）。尽管前景可观，其训练往往依赖昂贵的伴随法（Chen et al., 2018），难以扩展，且经常不稳定。相比之下，我们的 PT-Flow 直接使用近似真实潜在速度监督潜在 ODE 速度场，兼顾 ODE 性质的保持与高效、可扩展训练。

3 物理时间流

为超越离散时间建模，构建捕捉连续时间物理现实的新架构，我们引入物理时间流（PT-Flow）。核心思想是在恰当潜空间中，以连续速度场捕捉高维数据的底层动力学。令 $s_t \in \mathbb{R}^{d_s}$ 表示物理时刻 t的状态输入，$z_t \in \mathbb{R}^{d_z}$ 为 s_t经过适当编码后的潜在表征。 t 表示从初始状态 s_0开始后经过的物理时间。在本设置中， s_0 可灵活选为示范中的任意帧，而不限于首帧。因此， t定义为 s_0 与 s_t之间的相对时间偏移。为捕捉物理系统随时间演化的动力学，一种直接方法是学习以下常微分方程定义的时变速度场：

$$\frac{dz_t}{dt} = v_\theta(z_t, t; z_0, c) \implies z_T = z_0 + \int_0^T v_\theta(z_t, t; z_0, c)dt \tag{1}$$

其中 z_0 为初始状态 s_0 的潜在表征， c 为目标条件（如目标状态或语言指令），二者共同指定系统起止状态。若能学习潜在速度场 v_θ，未来预测就可重定义为在

the continuous physical time, thereby unlocking many interesting capabilities, such as arbitrary temporal resolution or backward prediction, as well as being amenable to many nice properties in control theory (Cellier and Greifeneder, 2013). The remaining questions are, what is the "proper" latent space and how can we learn v_θ with discrete-time supervision data?

Dynamical representation decoupling. Obviously, not all dynamical systems can be modeled as an ODE function, especially for nuanced mixtures of visual features in a video. Ideally, for video data, we do not want to learn a velocity field to fit every static background detail and time-invariant object, but want to extract clean, essential dynamics-related representations z_t out of the state space. To achieve this, we learn a pair of initial state-conditioned encoder and decoder: $f_{\text{dyn}}(s_t; s_0) = z_t$ and $g_{\text{dyn}}(z_t; s_0) = s_t$. By explicitly isolating static context (time-invariant part between s_0 and s_t) in both encoder and decoder, we make the encoded latent representation z_t focus exclusively on dynamical changes. To prevent information loss, we introduce the following reconstruction loss:

$$\mathcal{L}_{\text{dyn-recon}} = \mathbb{E}_{s_0,s_t\sim\mathcal{D}}\|g_{\text{dyn}}(f_{\text{dyn}}(s_t; s_0); s_0) - s_t\|^2. \tag{2}$$

To further encourage the encoder f_{dyn} to extract fundamental, ODE-compatible dynamical representations, we also enforce f_{dyn} to also satisfy the ODE property, i.e., $\dot{z}_t = \frac{df_{\text{dyn}}(s_t;s_0)}{dt}$. Next, we will show how this can facilitate the latent velocity field learning while achieving a direct, harmonious supervision on both f_{dyn} and v_θ.

Direct first-order supervision. Note that the time derivative of the latent state satisfies

$$\dot{z}_t = \frac{dz_t}{dt} = \frac{df_{\text{dyn}}(s_t; s_0)}{dt} = \frac{\partial f_{\text{dyn}}(s_t; s_0)}{\partial s_t}\cdot\frac{ds_t}{dt} + \cancel{\frac{\partial f_{\text{dyn}}(s_t; s_0)}{\partial s_0}\cdot\frac{ds_0}{dt}} = \frac{\partial f_{\text{dyn}}(s_t; s_0)}{\partial s_t}\cdot\dot{s}_t \tag{3}$$

As s_0 is time-invariant, hence $\frac{ds_0}{dt} = 0$ and the related term vanishes. The above equation suggests that $\dot{z}_t$ can be directly obtained utilizing a Jacobian-vector product (JVP) projection, i.e., $\dot{z}_t = \text{JVP}(f_{\text{dyn}}(s_t; s_0), \dot{s}_t)$. As $\dot{s}_t$ can be directly approximated from training samples, i.e., simply calculating as $\dot{s}_t = (s_{t+1} - s_t)/\Delta t$ (Δt is the time interval between two consecutive states) or adopting more advanced higher-order estimation tricks as we will describe in the next section, the evaluation of $\dot{z}_t$ is practically feasible. This enables us to introduce the following learning objective for the latent velocity field v_θ, which directly supervises it with the approximated ground-truth latent velocities $\dot{z}_t$:

$$\mathcal{L}_v = \mathbb{E}_{s_0,s_t\sim\mathcal{D},t}||v_\theta(z_t, t; z_0, c) - \text{sg}(\text{JVP}(f_{\text{dyn}}(s_t; s_0), \dot{s}_t))||^2, \tag{4}$$

We use the stop-gradient operation $\text{sg}(\cdot)$ to stabilize the optimization of the latent velocity field, following common practices in representation learning (Chen and He, 2021). Interestingly, we find that PT-Flow remains effective even without stop-gradient, as shown in App. B.6. This suggests that the proposed objective itself can provide effective constraints for learning meaningful representations. Unlike existing JEPA-style latent world models (Assran et al., 2023; Bardes et al., 2024; Assran et al., 2025) that supervise the latent predictor via the consistency loss between current and predicted embeddings, PT-Flow employs a direct first-order supervision strategy to specifically optimize the latent velocity field, effectively decoupling the dynamics learning from embedding reconstruction. Such a design greatly mitigates the risk of representation collapse (Wang et al., 2026a; Maes et al., 2026) in JEPA frameworks, and ensures the latent manifold remains expressive and dynamically meaningful.

By jointly optimizing the loss objectives $\mathcal{L}_{\text{dyn-recon}}$ and $\mathcal{L}_v$, we can learn a well-structured latent space and velocity field, in which the future prediction at an arbitrary time can be achieved by integrating with an off-the-shelf ODE solver in the compact latent space.

4 ODEWorld: Latent World Modeling via PT-Flow

In this section, we instantiate the PT-Flow paradigm as **ODEWorld** (illustrated in Fig. 1(c)), a highly versatile and efficient continuous-time predictive architecture for both high-fidelity video generation and effective robotic policy learning.

连续物理时间，从而解锁任意时间分辨率、反向预测等有趣能力，并兼容控制理论中许多良好性质（Cellier and Greifeneder, 2013）。剩下的问题是：什么是“恰当”的潜空间，如何用离散时间监督数据学习 v_θ？

动力学表征解耦。显然，并非所有动力系统都能建模为 ODE 函数，尤其是视频中细微视觉特征的混合。理想情况下，不应让速度场拟合每一项静态背景细节和时间不变物体，而应从状态空间提取干净、关键的动力学相关表征 z_t。为此，学习一对以初始状态为条件的编码器与解码器：$f_{\text{dyn}}(s_t; s_0) = z_t$ 和 $g_{\text{dyn}}(z_t; s_0) = s_t$。在编码器与解码器中显式分离静态上下文（$s_0$ 和 s_t之间时间不变的部分），使潜在表征 z_t 专注于动力学变化。为防止信息损失，引入以下重建损失：

$$\mathcal{L}_{\text{dyn-recon}} = \mathbb{E}_{s_0,s_t\sim\mathcal{D}}\|g_{\text{dyn}}(f_{\text{dyn}}(s_t; s_0); s_0) - s_t\|^2. \tag{2}$$

为进一步促使编码器 f_dyn 提取基本且兼容 ODE 的动力学表征，也要求 f_dyn 满足 ODE 性质，即 dz_t/dt = d f_dyn(s_t; s_0)/dt。接下来说明这如何促进潜在速度场学习，同时对 f_dyn 和 v_theta 提供直接、协调的监督。

直接一阶监督。注意，潜在状态的时间导数满足

$$\dot{z}_t = \frac{dz_t}{dt} = \frac{df_{\text{dyn}}(s_t; s_0)}{dt} = \frac{\partial f_{\text{dyn}}(s_t; s_0)}{\partial s_t}\cdot\frac{ds_t}{dt} + \cancel{\frac{\partial f_{\text{dyn}}(s_t; s_0)}{\partial s_0}\cdot\frac{ds_0}{dt}} = \frac{\partial f_{\text{dyn}}(s_t; s_0)}{\partial s_t}\cdot\dot{s}_t \tag{3}$$

由于 s_0 不随时间变化，因此 $\frac{ds_0}{dt} = 0$，相关项消失。上式说明，$\dot{z}_t$可通过雅可比—向量乘积（JVP）投影直接得到，即 $\dot{z}_t = \text{JVP}(f_{\text{dyn}}(s_t; s_0) \dot{s}_t)$。$\dot{s}_t$ 可直接由训练样本近似：简单计算为 $\dot{s}_t = (s_{t+1} - s_t)/\Delta t$（$\Delta t$ 是连续两个状态的时间间隔），或采用下一节介绍的更先进高阶估计方法，因此 $\dot{z}_t$ 的计算在实践中可行。由此可对潜在速度场 v_θ引入以下学习目标，以近似真实潜在速度 $\dot{z}_t$直接监督它：

$$\mathcal{L}_v = \mathbb{E}_{s_0,s_t\sim\mathcal{D},t}||v_\theta(z_t, t; z_0, c) - \text{sg}(\text{JVP}(f_{\text{dyn}}(s_t; s_0), \dot{s}_t))||^2, \tag{4}$$

遵循表征学习常见实践（Chen and He, 2021），使用停止梯度操作 sg（·）稳定潜在速度场优化。有趣的是，附录 B.6 表明，即使不停止梯度，PT-Flow 依然有效。这说明目标本身可为有意义表征的学习提供有效约束。现有 JEPA 风格潜在世界模型（Assran et al., 2023; Bardes et al., 2024; Assran et al., 2025）通过当前与预测嵌入的一致性损失监督潜在预测器；PT-Flow 则使用直接一阶监督专门优化潜在速度场，将动力学学习与嵌入重建有效解耦。这显著降低 JEPA 框架中的表征坍塌风险（Wang et al., 2026a; Maes et al., 2026），使潜在流形保持表达能力和动力学意义。

联合优化损失 $\mathcal{L}_{\text{dyn-recon}}$ 和 $\mathcal{L}_v$，即可学习结构良好的潜空间和速度场；通过在紧凑潜空间中用现成 ODE 求解器积分，实现任意时刻的未来预测。

4 ODEWorld：基于 PT-Flow 的潜在世界建模

本节将 PT-Flow 范式实例化为 ODEWorld（图 1(c)），一个高度通用、高效的连续时间预测架构，兼顾高保真视频生成和有效机器人策略学习。

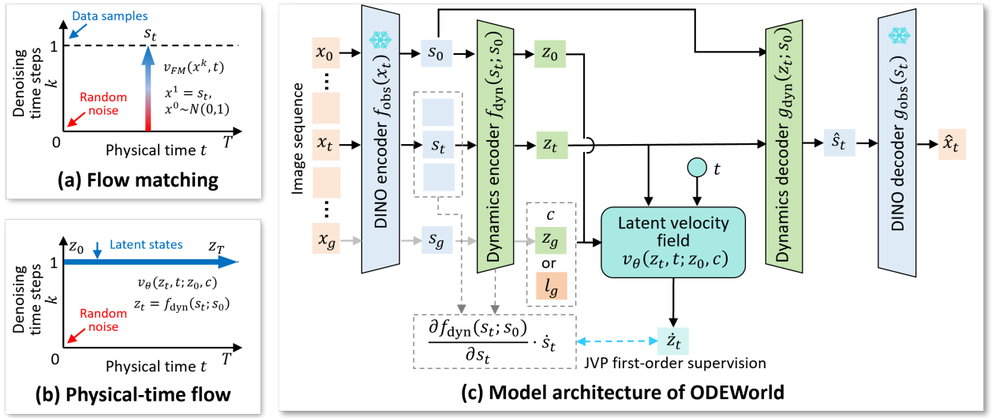


Figure 1 Illustration of PT-Flow and ODEWorld. (a), (b) Comparison between flow matching and our proposed PT-Flow. Although both methods learn time-dependent velocity fields and perform generation with ODE solvers, their notion of time and design philosophies are completely different. (c) Model architecture of ODEWorld.

Architecture overview. Apart from building PT-Flow, we first employ a frozen, pre-trained DINOv2 encoder (Oquab et al., 2024) as our vision backbone f_{obs} to project high-dimensional raw image observations x_t into a compressed feature space s_t. This transformed state space offers superior numerical stability compared to raw pixels when estimating the velocity target $\dot{s}_t$ in Eq. (3). Correspondingly, we train a dedicated image decoder g_{obs} to reconstruct the DINO feature back to image space $\hat{x}_t$ for video generation. We follow the same recipe with RAE (Zheng et al., 2026a) to train g_{obs} with a combination of L1, LPIPS (Zhang et al., 2018a) and adversarial losses (Goodfellow et al., 2014):

$$\begin{aligned} s_t = f_{obs}(x_t), \quad \hat{x}_t = g_{obs}(s_t), \quad x_t \in \mathcal{D} \\ \mathcal{L}_{\text{DINO-recon}}(x_t) = \mathcal{L}_1(\hat{x}_t, x_t) + \omega_L \mathcal{L}_{\text{LPIPS}}(\hat{x}_t, x_t) + \omega_G \lambda \mathcal{L}_{\text{GAN}}(\hat{x}_t, x_t). \end{aligned} \tag{5}$$

We further embed PT-Flow within the DINO feature space, where DINO image encoder and decoder f_{obs} and g_{obs} are frozen during the training of PT-Flow. We find that PT-Flow is sufficiently expressive to maintain consistency in latent planning and reconstruction within the DINO feature space, ultimately enabling high-fidelity video generation through g_{obs}.

Following the design of PT-Flow, we introduce an initial state-conditioned dynamics encoder f_{dyn} and a decoder g_{dyn}, both of which are implemented using cross-attention blocks. The dynamics encoder $f_{\text{dyn}}(s_t; s_0) = z_t$ extracts the dynamical representation z_t from the current state s_t conditioned on the initial state s_0, where a few learnable query tokens attend to both s_t and s_0 as keys and values, producing a compact latent representation that summarizes temporal changes between the initial and current states. The decoder g_{dyn} shares a similar structure, again with the initial state s_0 injected but serves as the query, while the latent representation z_{t} acts as keys and values. This design forces the model to reconstruct s_t by combining static context from s_0 with dynamical information encoded in z_t, thereby preventing z_t from carrying static content.

In our experiments, we find that a single token is sufficient to model z_t while achieving strong performance. Specifically, $z_t \in \mathbb{R}^{1\times 768}$ is highly compact compared to $s_t \in \mathbb{R}^{16\times 16\times 768}$ in original DINO space. More importantly, this result further supports the key insight of latent world modeling (Assran et al., 2023; Maes et al., 2026): raw pixels contain substantial redundancy, while the underlying dynamics can be effectively captured by a much smaller latent representation. Based on the compact z_t, we parameterize the velocity network $v_\theta(z_t, t; z_0, c)$ by a lightweight 3-layer MLP. The time t is incorporated by Feature-wise Linear Modulation (FiLM) layers (Perez et al., 2018), while other parts z_t, z_0, c are concatenated and fed into the model as input. Such a compact latent dynamics model design makes the latent planning and prediction with the ODE solver extremely efficient, while maximally preserving information for image reconstruction.

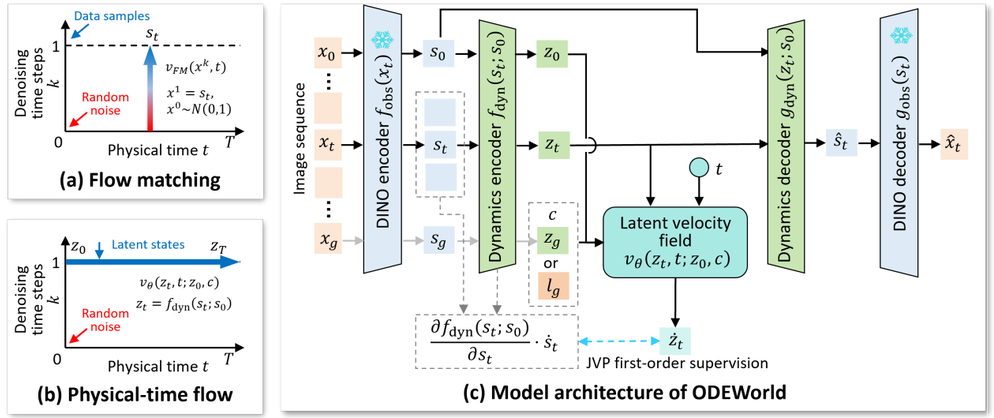


图 1：PT-Flow 与 ODEWorld 示意。（a）、（b）流匹配与 PT-Flow 的比较。两者都学习时变速度场并通过 ODE 求解器生成，但时间概念和设计理念完全不同。（c）ODEWorld 模型架构。

架构概览。除构建 PT-Flow 外，首先采用冻结的预训练 DINOv2 编码器（Oquab et al., 2024）作为视觉骨干 f_{obs}，将高维原始图像观测 x_t投影到压缩特征空间 s_t。相较原始像素，这种转换后的状态空间在估计式（3）的速度目标 $\dot{s}_t$ 时，数值稳定性更好。相应地，训练专用图像解码器 g_{obs}，将 DINO 特征重建回图像空间 $\hat{x}_t$，用于视频生成。遵循 RAE（Zheng et al., 2026a）的相同方案，结合 L1、LPIPS（Zhang et al., 2018a）及对抗损失（Goodfellow et al., 2014）训练 g_{obs}：

$$\begin{aligned} s_t = f_{obs}(x_t), \quad \hat{x}_t = g_{obs}(s_t), \quad x_t \in \mathcal{D} \\ \mathcal{L}_{\text{DINO-recon}}(x_t) = \mathcal{L}_1(\hat{x}_t, x_t) + \omega_L \mathcal{L}_{\text{LPIPS}}(\hat{x}_t, x_t) + \omega_G \lambda \mathcal{L}_{\text{GAN}}(\hat{x}_t, x_t). \end{aligned} \tag{5}$$

进一步将 PT-Flow 嵌入 DINO 特征空间；训练 PT-Flow 时冻结 DINO 图像编码器和解码器 f_{obs}及 g_{obs}。PT-Flow 的表达能力足以保持 DINO 特征空间中潜在规划与重建的一致性，最终通过 g_{obs}实现高保真视频生成。

遵循 PT-Flow 设计，引入以初始状态为条件的动力学编码器 f_{dyn} 和解码器 g_{dyn}，均通过交叉注意力块实现。动力学编码器 f_{dyn}（s_t；s_0）$= z_t$以初始状态 s_0为条件，从当前状态 s_t 提取动力学表征 z_t；少量可学习查询标记同时关注作为键和值的 s_t 及 s_0，输出概括初始与当前状态间时间变化的紧凑潜在表征。解码器 g_{dyn}结构类似，也注入初始状态 s_0，但使其作为查询，潜在表征 z_{t} 作为键和值。这迫使模型结合来自 s_0 的静态上下文和 z_t编码的动力学信息重建 s_t，从而避免 z_t 携带静态内容。

实验发现，单个标记足以建模 z_t 并取得较强性能。具体而言，相比原始 DINO 空间的 $s_t \in \mathbb{R}^{16\times 16\times 768}$，$z_t \in \mathbb{R}^{1\times 768}$ 高度紧凑。更重要的是，这进一步支持潜在世界建模的核心见解（Assran et al., 2023; Maes et al., 2026）：原始像素含大量冗余，而底层动力学可由更小的潜在表征有效捕捉。基于紧凑的 z_t，我们用轻量三层 MLP 参数化速度网络 v_θ（z_t、t；z_0、c）。时间 t通过特征级线性调制（FiLM）层（Perez et al., 2018）引入，其余 z_t、z_0、c 拼接后作为模型输入。紧凑的潜在动力学设计使 ODE 求解器的潜在规划与预测极为高效，同时最大限度保留图像重建信息。

Additional enhancements on PT-Flow. To ensure numerical stability, instead of modeling the latent velocity $v_\theta(z_t, t; z_0, c)$ with the actual physical time t as in Eq. (1), we implement $v_\theta(z_t, \tau; z_0, c)$ based on a rescaled physical time τ, where $\tau = \frac{t}{L}$ and L is a normalizing horizon length which is approximately the desired planning length. This will make sure τ mostly resides within the range $[0, 1]$ during training. Under this reparameterization, $\dot{s}_\tau$ corresponds to $\dot{s}_t$ scaled by L in time dimension. We assume that the system reaches a stationary state once the target condition c is reached (e.g., the goal state z_g or language instruction l_g). Therefore, when τ exceeds the rescaled time index τ_c of the last frame in a training sequence ending with condition c, we fix the future state $s_{\tau>\tau_c}$ to the last frame and drive the velocity field v_θ to zero in the end.

To enhance the estimation quality of the target velocity $\dot{s}_t$ for JVP projection in Eq. (4), we employ a Savitzky–Golay derivative filter (Saramäki et al., 1993), which performs local polynomial regression over a sliding window. Specifically, given a sequence of states $\{s_{t-k}, \ldots, s_{t+k}\}$ within a window of size $2k+1$, the target velocity is calculated as a linear convolution: $\dot{s}_t = \sum_{i=-k}^{k} w_i s_{t+i}$, where w_i are fixed coefficients obtained by fitting a low-order polynomial in a least-squares sense. In our implementation, we instantiate this filter with a window size of $2k+1 = 5$, yielding a filter kernel $w = \frac{1}{10}[-2, -1, 0, 1, 2]$. Unlike conventional smoothing kernels that blindly average signals (e.g., Gaussian blurring) or standard finite difference kernels, the chosen Savitzky–Golay kernel provides a decent compromise between numerical accuracy and temporal smoothness, due to its adoption of local least-squares polynomial fit. This ensures that the projected target $\dot{z}_t$ (via Eq. (3)) is both temporally smooth and physically accurate, which substantially enhances the learning performance of the latent velocity field v_θ.

Training and inference pipeline. With the pre-trained frozen DINO encoder f_{obs} and decoder g_{obs}, the training of ODEWorld strictly follows the recipe of PT-Flow described in Section 3, which jointly minimizes the dynamics reconstruction loss $\mathcal{L}_{\text{dyn-recon}}$ (Eq. 2) and the velocity supervision loss $\mathcal{L}_v$ (Eq. 4), i.e.,

$$\mathcal{L} = \lambda_{rec}\mathcal{L}_{\text{dyn-recon}} + \lambda_v \mathcal{L}_v \tag{6}$$

In our experiments, we find that simply setting the weighting parameters $\lambda_{rec} = \lambda_v = 1$ without any tuning generally leads to reasonably good results. This further demonstrates the learning robustness of our proposed PT-Flow framework.

During inference, ODEWorld serves as a continuous-time ODE engine for both high-fidelity video synthesis and efficient latent planning. The inference pipeline works as follows: given an initial raw observation x_0 and a target condition c, the model first projects the observation into the DINO space $s_0 = f_{obs}(x_0)$ and then encodes this state into the dynamical latent space $z_0 = f_{dyn}(s_0; s_0)$. We can employ any off-the-shelf ODE solver, e.g. fourth-order Runge-Kutta method (RK4), to integrate the learned velocity field v_θ by $z_\tau = z_0 + \int_0^\tau v_\theta(z_t, t; z_0, c)dt$ to predict the latent state z_τ at arbitrary time $\tau \in [0, 1]$. The evolved latent state z_τ can then be used for: 1) future video frame generation: the latent state is sequentially mapped back to the pixel space with $\hat{s}_\tau = g_{dyn}(z_\tau; s_0)$ and $\hat{x}_\tau = g_{obs}(\hat{s}_\tau)$; and 2) downstream policy learning: z_τ serves as a compact, dynamically meaningful subgoal. Its low-dimensional dynamics-centric nature provides a stable target for policy learning, bypassing the complexity of high-dimensional visual planning. A notable advantage of ODEWorld lies in its remarkable inference speed. By operating within the compact latent manifold z and utilizing a shallow MLP to parameterize the velocity field v_θ, the numerical integration becomes computationally lightweight. This efficiency allows for real-time video generation and fast iterative latent planning, facilitating responsive closed-loop control in complex environments.

5 Experiments

5.1 Experimental Setups

Datasets. We train ODEWorld on LIBERO (Liu et al., 2023) simulation dataset and AgiBot-World (Bu et al., 2025) real-world robot datasets. LIBERO consists of 130 tasks with 6,500 high-quality human teleoperation demonstrations, covering diverse motions and atomic skills. We use the full dataset to train ODEWorld and all baseline models for image generation. For the AgiBot-World, we train ODEWorld on the AgiBotWorldChallenge-2025 subset, a large-scale embodied manipulation benchmark built upon the AgiBot World Colosseo platform (Bu et al., 2025). The dataset contains approximately 30K manipulation

PT-Flow 的额外改进。为确保数值稳定，我们不按式（1）直接使用实际物理时间 t 建模潜在速度 v_θ（z_t、t；z_0、c），而基于缩放后的物理时间 τ 实现 v_θ（z_t、τ；z_0、c），其中 $\tau = \frac{t}{L}$，L 为约等于所需规划长度的归一化时域长度。这使训练期间 τ 主要位于 $[0, 1]$ 范围。此重参数化下，$\dot{s}_\tau$ 对应于在时间维度按 L 缩放的 $\dot{s}_t$。我们假定达到目标条件 c（例如目标状态 z_g 或语言指令 l_g）后，系统进入静止状态。因此，当 τ 超过以条件 c结束的训练序列最后一帧的缩放时间索引 τ_c 时，将未来状态 $s_{\tau>\tau_c}$ 固定为最后一帧，并最终使速度场 v_θ 趋于零。

为提升式（4）中用于 JVP 投影的目标速度 $\dot{s}_t$ 估计质量，我们采用 Savitzky–Golay 导数滤波器（Saramäki et al., 1993），在滑动窗口内进行局部多项式回归。具体而言，给定大小为 $2k+1$的窗口内的状态序列 $\{s_{t-k}, \ldots, s_{t+k}\}$，目标速度以线性卷积计算：$\dot{s}_t = \sum_{i=-k}^{k} w_i s_{t+i}$，其中 w_i 为按最小二乘拟合低阶多项式得到的固定系数。实现中使用窗口大小 $2k+1 = 5$，得到滤波核 $w = \frac{1}{10}[-2, -1, 0, 1, 2]$。相比盲目平均信号的传统平滑核（如高斯模糊）或标准有限差分核，所选 Savitzky–Golay 核通过局部最小二乘多项式拟合，在数值精度与时间平滑性之间取得良好折中。这确保经式（3）投影的目标 $\dot{z}_t$ 同时具有时间平滑性与物理准确性，显著提升潜在速度场 v_θ的学习效果。

训练与推断流水线。使用预训练并冻结的 DINO 编码器 f_{obs} 和解码器 g_{obs}，ODEWorld 严格遵循第 3 节的 PT-Flow 方案，联合最小化动力学重建损失 $\mathcal{L}_{\text{dyn-recon}}$（式 2）和速度监督损失 $\mathcal{L}_v$（式 4），即

$$\mathcal{L} = \lambda_{rec}\mathcal{L}_{\text{dyn-recon}} + \lambda_v \mathcal{L}_v \tag{6}$$

实验中，直接设置权重参数 $\lambda_{rec} = \lambda_v = 1$，无需调参，通常就能取得较好结果。这进一步体现 PT-Flow 框架学习的鲁棒性。

推断时，ODEWorld 作为连续时间 ODE 引擎，同时支持高保真视频合成与高效潜在规划。流程如下：给定初始原始观测 x_0及目标条件 c，模型先将观测投影到 DINO 空间 $s_0 = f_{obs}(x_0)$，再将该状态编码到动力学潜空间 $z_0 = f_{dyn}(s_0; s_0)$。可以使用任意现成 ODE 求解器，如四阶 Runge–Kutta 方法（RK4），按照 $z_\tau = z_0 + \int_0^\tau v_\theta(z_t, t; z_0, c)\,dt$ 对学得的速度场 v_θ 积分，以预测任意时刻 $\tau \in [0, 1]$的潜在状态 z_τ。演化后的潜在状态 z_τ 可用于：1）未来视频帧生成：通过$\hat{s}_\tau = g_{dyn}(z_\tau; s_0)$ 及 $\hat{x}_\tau = g_{obs}(\hat{s}_\tau)$ 依次映射回像素空间；2）下游策略学习：z_τ作为紧凑、具有动力学意义的子目标。它低维且以动力学为中心，为策略学习提供稳定目标，绕开高维视觉规划的复杂性。ODEWorld 的显著优势是推断速度。由于在紧凑潜在流形 z 中运行，并以浅层 MLP 参数化速度场 v_θ，数值积分的计算量很小。这种效率支持实时视频生成与快速迭代的潜在规划，使复杂环境中的闭环控制响应及时。

5 实验

5.1 实验设置

数据集。在 LIBERO（Liu et al., 2023）模拟数据集和 AgiBot-World（Bu et al., 2025）真实机器人数据集上训练 ODEWorld。LIBERO 包含 130 项任务、6,500 条高质量人类遥操作示范，覆盖多样运动和原子技能。使用完整数据集训练 ODEWorld 和全部图像生成基线。对于 AgiBot-World，使用 AgiBotWorldChallenge-2025 子集，这是基于 AgiBot World Colosseo 平台（Bu et al., 2025）构建的大型具身操作基准。数据集包含约 3 万条操作

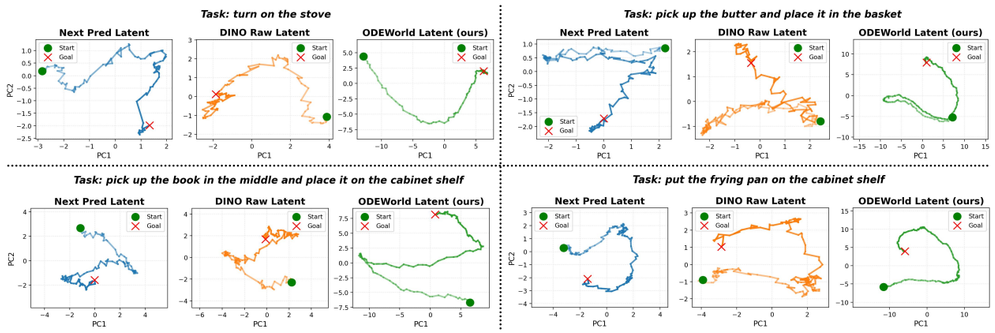


Figure 2 Data trajectories in different latent spaces. ODEWorld produces significantly smoother trajectories in its dynamical latent space across various LIBERO tasks compared to the latent spaces of DINO and the next-step prediction baseline.

trajectories with an average length of around 260 frames per trajectory, providing diverse robot-object interaction behaviors with visual observations across various real-world manipulation tasks. More details about datasets are provided in App. A.1.

Baselines. To evaluate the video generation quality of ODEWorld, we compare it against latent world models that also support reconstruction from semantic representations. Specifically, we consider Latent Diffusion Planning (LDP) (Xie et al., 2025) and V-JEPA 2 (Assran et al., 2025). LDP employs a reconstruction-based VAE latent space and performs planning and policy learning within this latent space, with generated latents decoded back into images via a VAE decoder. V-JEPA 2 jointly learns a representation encoder and a latent predictor, and similarly decodes predicted latents into raw images through a learned decoder. These models serve as suitable baselines for assessing generative quality in latent world model settings. As for policy learning, we compare ODEWorld against planning-based baselines, including the editing-based subgoal planner SuSIE (Black et al., 2024) and the end-to-end predictive inverse dynamics model Seer (Tian et al., 2025), the video prediction policy (Hu et al., 2025), and a vanilla goal-and-language-conditioned behavior cloning (GLCBC) policy. More details about these baselines are provided in App. A.3 and App. A.4.

5.2 Visualization

Latent space analysis. In Fig. 2, we compare the latent space of ODEWorld against the raw DINO feature space and the latent of a discrete next-step predictive baseline. For the latter, we replace v_θ with a discrete transition model $z_{t+1} = p_\theta(z_t; c)$, remove dynamical representation decoupling designs in f_{dyn} and g_{dyn}, and then optimize p_θ with JEPA-style objectives (Assran et al., 2023). PCA visualizations for latent trajectories from multiple LIBERO tasks reveal that ODEWorld yields the smoothest latent trajectories. This is attributed to our direct first-order supervision, which regularizes the manifold by aligning it with an analytically filtered velocity target that effectively suppresses high-frequency jitters. Crucially, while ODEWorld enhances dynamical smoothness, it maintains a manifold topology remarkably similar to the original DINO space. Unlike over-regularized models that tend to straighten the latent space to achieve smoothness at the expense of expressivity, ODEWorld's latent space preserves the principal components and the relative geometric structure in the original space. This structural fidelity ensures that the learned latent space retains rich semantic information, providing a high-fidelity foundation for downstream tasks such as video generation and subgoal planning for policy learning.

Velocity field and open-loop latent planning. In the middle row of Fig. 3, we sample a uniform grid in the projected PCA space and utilize the learned velocity field v_θ to compute the instantaneous latent velocities at each point. These are visualized as a vector field with arrows, the color of which represents the magnitude of dynamical transition. Starting from a latent state at a random time of the sampled trajectory, we perform a long-horizon rollout using an ODE solver (RK4) to integrate the velocity v_θ along normalized

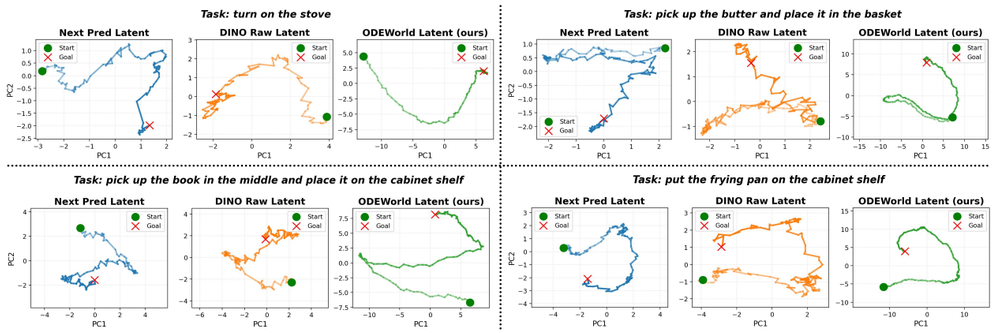


图 2：不同潜空间中的数据轨迹。相比 DINO 和下一步预测基线，ODEWorld 在各 LIBERO 任务的动力学潜空间中产生明显更平滑的轨迹。

轨迹，每条平均约 260 帧，提供多样机器人—物体交互行为及各类真实操作任务的视觉观测。数据集详见附录 A.1。

基线。为评估 ODEWorld 的视频生成质量，我们与同样支持从语义表征重建的潜在世界模型比较，具体包括潜在扩散规划（LDP）（Xie et al., 2025）和 V-JEPA 2（Assran et al., 2025）。LDP 使用基于重建的 VAE 潜空间，在其中开展规划和策略学习，并通过 VAE 解码器将生成潜变量解码为图像。V-JEPA 2 联合学习表征编码器和潜在预测器，也用学习到的解码器将预测潜变量还原为原始图像。这些模型适合评估潜在世界模型中的生成质量。对于策略学习，我们比较的规划基线包括基于编辑的子目标规划器 SuSIE（Black et al., 2024）、端到端预测逆动力学模型 Seer（Tian et al., 2025）、视频预测策略（Hu et al., 2025），以及普通目标与语言条件行为克隆（GLCBC）策略。详见附录 A.3 和 A.4。

5.2 可视化

潜空间分析。图 2 将 ODEWorld 潜空间与原始 DINO 特征空间及离散下一步预测基线的潜空间比较。对于后者，将 v_θ 替换为离散转移模型 $z_{t+1} = p_\theta(z_t;\ c)$，移除 f_{dyn} 和 g_{dyn}中的动力学表征解耦设计，再用 JEPA 风格目标优化 p_θ （Assran et al., 2023）。多个 LIBERO 任务潜在轨迹的 PCA 可视化表明，ODEWorld 的轨迹最平滑。这得益于直接一阶监督：通过使流形与解析滤波得到的速度目标对齐，抑制高频抖动。关键在于，ODEWorld 增强动力学平滑性的同时，流形拓扑仍与原始 DINO 空间高度相似。过度正则化的模型往往拉直潜空间以换取平滑，并损失表达能力；ODEWorld 则保留原空间的主成分及相对几何结构。这种结构保真确保潜空间保留丰富语义，为视频生成和策略学习子目标规划等下游任务提供高保真基础。

速度场与开环潜在规划。图 3 中排在投影后的 PCA 空间采样均匀网格，用学得的速度场 v_θ 计算各点瞬时潜在速度，并以箭头向量场可视化，颜色代表动力学转移幅值。从采样轨迹的随机时刻潜在状态出发，使用 ODE 求解器（RK4）沿归一化时间对速度 v_θ 积分，进行长时域生成，

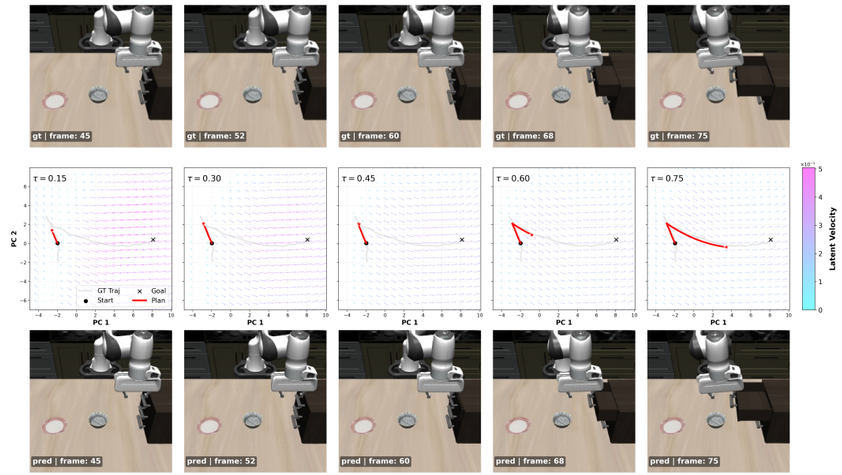


Figure 3 Velocity field and latent planning. The middle row shows visualization of globally consistent goal-directed convergence of velocity field and smooth latent plans from ODEWorld that adhere well to the ground truth. The top row shows ground truth frames, and the bottom row shows synthesized frames.

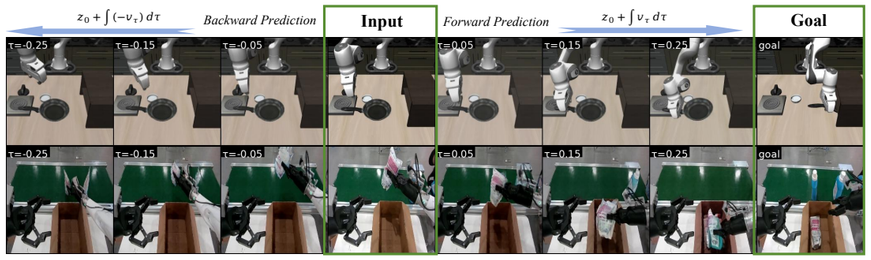


Figure 4 The continuous formulation enables ODEWorld to perform both forward and backward prediction, with motion consistent with real-world dynamics. We present the results on LIBERO benchmark and Agibot World. More results are provided in the Appendix.

time τ without any further environmental observations. To verify semantic fidelity, the integrated latent trajectory z_τ is passed through the decoders to synthesize image-space predictions shown in the bottom row, which are then compared against the ground-truth video sequence shown in the top row. The visualization reveals three key properties: (1) The velocity field captures the time-varying magnitude of the dynamics: in this example, the robot first approaches the handle and then pulls the drawer out, and the velocity correctly decays toward zero as the robot reaches the handle before accelerating again into the pulling phase. This shows that the learned velocity field encodes not only the direction toward the goal, but also the correct instantaneous speed at each stage of the task. (2) The model demonstrates high physical consistency even during open-loop planning. The planned trajectory (red line) adheres closely to the ground-truth latent path (gray line), while effectively filtering out the high-frequency noise and jitter present in the latter. (3) Finally, the synthesized predictions in the bottom row remain spatially coherent and semantically accurate. This combination of planning accuracy and smoothness provides a stable, jerk-free foundation for downstream high-fidelity video generation and precise subgoal planning for policy learning.

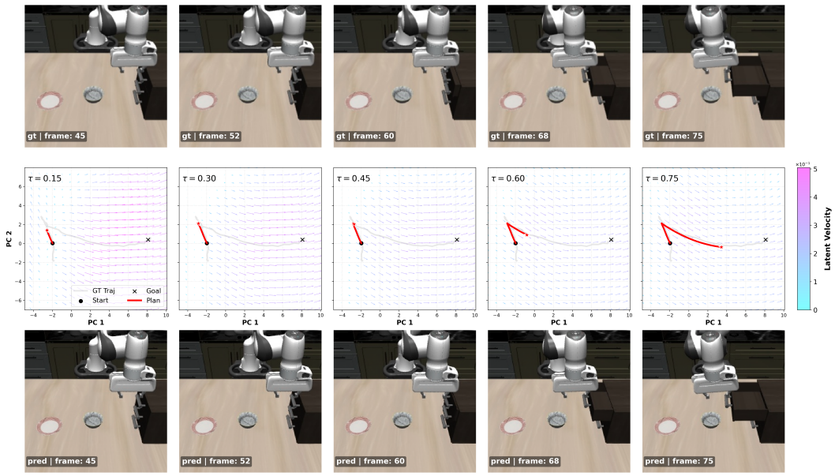


图 3：速度场与潜在规划。中排可视化显示，ODEWorld 速度场在全局一致地朝目标收敛，并产生紧贴真实值的平滑潜在规划。上排为真实帧，下排为合成帧。

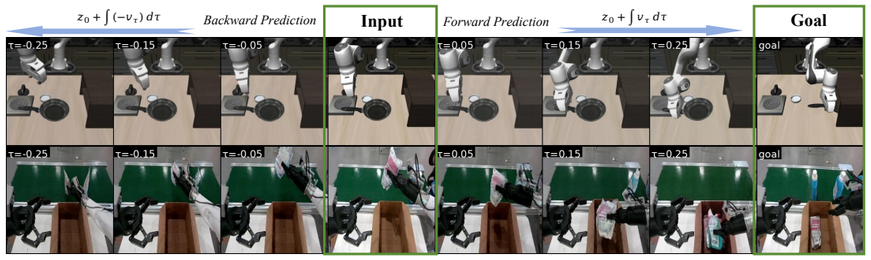


图 4：连续形式使 ODEWorld 同时实现前向和反向预测，运动与真实动力学一致。展示 LIBERO 基准及 Agibot World 的结果；更多结果见附录。

时刻 τ，不再接收任何环境观测。为验证语义保真度，将积分得到的潜在轨迹 z_τ 通过解码器合成为下排的图像空间预测，再与上排真实视频序列比较。可视化揭示三项性质：（1）速度场捕捉动力学幅值随时间变化：机器人先接近把手，再拉开抽屉；到达把手前，速度正确衰减至零，进入拉动阶段后再次加速。这表明速度场不仅编码朝向目标的方向，也编码任务各阶段正确的瞬时速率。（2）模型即使在开环规划时仍保持较高物理一致性。规划轨迹（红线）紧贴真实潜在路径（灰线），并有效滤除后者的高频噪声和抖动。（3）下排合成预测保持空间连贯和语义准确。规划精度与平滑性相结合，为下游高保真视频生成及策略学习中的精确子目标规划提供稳定、无突变抖动的基础。



Figure 5 ODEWorld allows any resolution generation with flexible temporal speed. Based on this, ODEWorld can generalize beyond the sampling rate of the training data. In particular, when trained on temporally downsampled sequences, the model is able to generate smooth and coherent intermediate frames, performing temporal super-resolution and filling in missing observations.

5.3 ODEWorld for Video Generation

Bidirectional video generation. As demonstrated in Fig. 4, ODEWorld exhibits unique bidirectional flexibility inherent to its ODE-governed architecture. By simply reversing the integration sign ($v_\tau \to -v_\tau$), the model can synthesize physically plausible backward video sequences from the same learned latent velocity field. This symmetry validates that ODEWorld internalizes the underlying continuous physics , rather than merely memorizing forward predictive patterns typical of those discrete predictive architectures.

Any-resolution video generation. As illustrated in Fig. 5, ODEWorld leverages its continuous, ODE-governed nature to achieve arbitrary temporal resolution in video generation. Unlike discrete models constrained by fixed step sizes, our model can generate motion at any speed (Fig. 5(a)) and effectively interpolate missing frames (Fig. 5(b)). As shown in Fig. 5(b), even when trained on a down-sampled dataset (e.g., 1/3 of the original frame rate), ODEWorld successfully recovers the intermediate temporal dynamics, a



图 5：ODEWorld 支持任意时间分辨率与灵活运动速度，因而可泛化到训练采样率之外。特别是用时间降采样序列训练后，仍能生成平滑、连贯的中间帧，实现时间超分辨率并填补缺失观测。

5.3 ODEWorld 视频生成

双向视频生成。图 4 表明，ODEWorld 具有 ODE 架构固有的双向灵活性。只需反转积分符号（$v_\tau \to -v_\tau$），即可从同一潜在速度场合成物理上合理的反向视频序列。此对称性验证 ODEWorld 内化了底层连续物理，而非仅记忆离散预测架构典型的前向预测模式。

任意时间分辨率视频生成。图 5 显示，ODEWorld 利用由 ODE 支配的连续性，实现任意时间分辨率的视频生成。离散模型受固定步长限制，而本模型可生成任意速度的运动（图 5(a)），并有效插值缺失帧（图 5(b)）。即使使用降采样数据训练（如原始帧率的 1/3），ODEWorld 仍能恢复中间时间动力学，这是一项

Table 1 Comparison on video prediction quality and latency.

Methods	Short Fragment @16frames			Long Fragment @64frames		
	PSNR↑	LPIPS↓	Latency(s)↓	PSNR↑	LPIPS↓	Latency(s)↓
LDP	16.33	0.489	1.104	16.27	0.461	3.953
V-JEPA 2	17.60	0.157	0.123	16.47	0.166	0.619
ODEWorld (Ours)	**20.53**	**0.109**	**0.030**	**19.46**	**0.134**	**0.072**

Table 2 On the LIBERO-LONG benchmark, ODEWorld achieves consistent improvements in policy performance. Averaged on 3 random seeds. Tasks are specified in App. A.1.

Method \ Task ID	1	2	3	4	5	6	7	8	9	10	Avg. Suc ↑
GLCBC	83.3	93.3	83.3	93.3	76.6	63.3	73.3	83.3	50.0	80.0	78.0
SuSIE	83.3	63.3	**96.6**	**100.0**	83.3	83.3	83.3	39.9	53.3	76.6	76.3
Seer	88.3	90.0	91.6	81.6	85.0	65.0	**86.6**	80.0	51.6	66.6	78.6
VPP	**96.7**	**100.0**	73.3	93.3	80.0	**96.7**	50.0	76.7	**70.0**	73.3	81.0
ODEWorld (Velocity)	80.0	93.3	76.6	**100.0**	73.3	80.0	83.3	86.6	66.6	83.3	82.3
ODEWorld (Single subgoal)	86.6	86.6	86.6	96.6	**86.6**	86.6	80.0	83.3	46.6	**86.6**	82.6
ODEWorld (Sequential subgoals)	90.0	96.6	90.0	**100.0**	73.3	90.0	76.6	**90.0**	60.0	80.0	**83.6**

Table 3 Real-world robot experiments with ODEWorld-guided policy learning.

Method	Shrimp	Mouse	Pen	Rearrangement	Average
X-VLA	80%	60%	30%	50%	55%
ODEWorld (Sequential subgoals)	**90%**	**80%**	**70%**	**80%**	**80%**

capability inherently absent in discrete-time baselines.

Quantitative comparisons. Tab. 1 provides a comprehensive evaluation of video prediction quality and efficiency. ODEWorld consistently outperforms baselines in both pixel-level metrics PSNR (Hore and Ziou, 2010) and perception-level metrics LPIPS (Zhang et al., 2018b) across short and long horizons. ODEWorld maintains high structural fidelity and perceptual realism even in long-horizon predictions (@64 frames). Notably, our model achieves substantially lower prediction latency (0.072s for @64 frames) on a single A100 GPU. These results demonstrate that ODEWorld provides accurate long-horizon predictions while maintaining an efficient architecture.

5.4 ODEWorld for Policy Learning

We assess how ODEWorld can empower downstream policy learning by inferring latent subgoals as guidance. We instantiate three guidance paradigms: (1) velocity-conditioned $\pi(a|z, c, v)$, (2) single-subgoal $\pi(a|z, c, \hat{z}_\tau), \tau = 0.25$, and (3) sequential-subgoal $\pi(a|z, c, \{\hat{z}_{\tau_i}\}_{i=1}^n), n = 5, \tau_i = 0.05i$, where subgoals are generated via ODE integration. c denotes image goal and language instruction.

Results on LIBERO-LONG. As shown in Tab. 2, all ODEWorld variants outperform the baselines on LIBERO-LONG. Among them, the sequential-subgoal paradigm achieves the best performance with an average success rate of 83.6%. This performance gain demonstrates the high-fidelity temporal consistency of ODEWorld rollouts; by maintaining precise trajectory alignment over horizon, the model provides dense and reliable guidance that does not drift from the task objective. Such consistent sequential subgoal guidance effectively benefits downstream policy learning.

Results on real-world robot experiments. To further validate the effectiveness of ODEWorld beyond simulation benchmarks, we conduct real-world control experiments on a bi-manual AgileX robot. We use X-VLA (Zheng et al., 2026b) as the policy backbone and augment it with ODEWorld-generated latent subgoals for guidance. We evaluate four manipulation tasks, including: *Shrimp-to-pot*, *Mouse packing*, *Pen insertion*,

表 1：视频预测质量与延迟比较。

方法	短片段：16 帧			长片段：64 帧		
	PSNR↑	LPIPS↓	延迟（秒）↓	PSNR↑	LPIPS↓	延迟（秒）↓
LDP	16.33	0.489	1.104	16.27	0.461	3.953
V-JEPA 2	17.60	0.157	0.123	16.47	0.166	0.619
ODEWorld（本文）	**20.53**	**0.109**	**0.030**	**19.46**	**0.134**	**0.072**

表 2：ODEWorld 在 LIBERO-LONG 基准上持续改善策略性能。结果对 3 个随机种子取平均。任务见附录 A.1。

方法 \ 任务 ID	1	2	3	4	5	6	7	8	9	10	平均成功率 ↑
GLCBC	83.3	93.3	83.3	93.3	76.6	63.3	73.3	83.3	50.0	80.0	78.0
SuSIE	83.3	63.3	**96.6**	**100.0**	83.3	83.3	83.3	39.9	53.3	76.6	76.3
Seer	88.3	90.0	91.6	81.6	85.0	65.0	**86.6**	80.0	51.6	66.6	78.6
VPP	**96.7**	**100.0**	73.3	93.3	80.0	**96.7**	50.0	76.7	**70.0**	73.3	81.0
ODEWorld（速度）	80.0	93.3	76.6	**100.0**	73.3	80.0	83.3	86.6	66.6	83.3	82.3
ODEWorld（单子目标）	86.6	86.6	86.6	96.6	**86.6**	86.6	80.0	83.3	46.6	**86.6**	82.6
ODEWorld（顺序子目标）	90.0	96.6	90.0	**100.0**	73.3	90.0	76.6	**90.0**	60.0	80.0	**83.6**

表 3：ODEWorld 引导策略学习的真实机器人实验。

方法	虾	鼠标	Pen	重排 平均值	
X-VLA	80%	60%	30%	50%	55%
ODEWorld（顺序子目标）	**90%**	**80%**	**70%**	**80%**	**80%**

离散时间基线固有地不具备的能力。

定量比较。表 1 全面评估视频预测质量与效率。在短、长时域上，ODEWorld 的像素级 PSNR（Hore and Ziou, 2010）和感知级 LPIPS（Zhang et al., 2018b）均持续优于基线。即使预测 64 帧的长时域，仍保持较高结构保真度与感知逼真度。值得关注的是，在单张 A100 GPU 上，模型预测延迟明显更低（预测 64 帧为 0.072 秒）。结果表明，ODEWorld 以高效架构实现准确的长时域预测。

5.4 ODEWorld 策略学习

评估 ODEWorld 如何推断潜在子目标作为引导，增强下游策略学习。实现三种引导范式：（1）速度条件 π（$a|z$、c、v）；（2）单子目标 π（$a|z$、c、$\hat{z}_\tau$） $\tau = 0.25$；（3）顺序子目标 π（$a|z$、c、$\{\hat{z}_{\tau_i}\}_{i=1}^n$） $n = 5$、$\tau_i = 0.05i$，其中子目标通过 ODE 积分生成。 c 表示图像目标与语言指令。

LIBERO-LONG 结果。表 2 中，所有 ODEWorld 变体均优于基线。其中顺序子目标范式表现最好，平均成功率 83.6%。提升体现了 ODEWorld 生成轨迹的高保真时间一致性：通过在整个时域保持精确轨迹对齐，模型提供密集、可靠且不偏离任务目标的引导。这样的顺序子目标引导有效促进下游策略学习。

真实机器人实验结果。为在模拟基准之外验证有效性，在 AgileX 双臂机器人上开展真实控制实验。采用 X-VLA（Zheng et al., 2026b）作为策略骨干，以 ODEWorld 生成的潜在子目标增强引导。评估四项操作任务，包括虾入锅、鼠标装盒、插笔，

Bi-manual rearrangement. Detailed task settings are provided in App. A.2. Using the same policy backbone and training protocol, ODEWorld improves the average success rate from 55% to 80% across these four tasks, demonstrating the effectiveness of ODEWorld-generated dynamics guidance in real-world manipulation scenarios.

6 Conclusions

We introduce ODEWorld, a novel continuous latent world model grounded in physical time. The core of ODEWorld is a new predictive paradigm called PT-Flow, which learns a latent ODE velocity field defined on physical time. It recasts future prediction as temporal integration with the latent velocity via an ODE solver, which shares some similarities to flow matching, but operates in physical time rather than the noise space. Through comprehensive experiments on both simulation and real-world benchmark datasets, we show that ODEWorld enjoys a number of desirable properties, such as interpretable latent velocity fields, prediction with arbitrary directions or temporal resolutions, high-quality reconstruction after long-term prediction, and planning-centric representations that facilitate downstream policy learning. Many of the above properties are fundamentally absent in most discrete-time predictive models. We anticipate that PT-Flow and ODEWorld will inspire a new generation of predictive models, paving the way for a more faithful understanding and modeling of the continuous physical world.

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双臂重排。详细任务设置见附录 A.2。使用相同策略骨干和训练协议，ODEWorld 将四项任务的平均成功率从 55% 提升到 80%，证明其生成的动力学引导在真实操作场景中有效。

6 结论

我们提出 ODEWorld，一个基于物理时间的新型连续潜在世界模型。核心是名为 PT-Flow 的新预测范式，学习定义于物理时间的潜在 ODE 速度场，将未来预测重述为通过 ODE 求解器对潜在速度进行时间积分。它与流匹配有相似之处，但运行在物理时间而非噪声空间。通过模拟与真实世界基准上的全面实验，我们表明 ODEWorld 具有可解释的潜在速度场、任意方向或时间分辨率预测、长期预测后的高质量重建，以及促进下游策略学习的规划中心表征等性质。多数离散时间预测模型根本不具备其中许多性质。我们期待 PT-Flow 与 ODEWorld 激发新一代预测模型，为更忠实地理解和建模连续物理世界铺路。

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Appendix

A Experiment Details

A.1 Datasets

LIBERO (Liu et al., 2023). LIBERO is a large-scale language-conditioned robotic manipulation benchmark comprising 130 tasks and 6,500 high-quality human teleoperation demonstrations. These tasks span a diverse set of atomic and compositional skills, including grasping, lifting, placing, opening, and multi-step object rearrangement. The benchmark is organized into five suites, namely LIBERO-Spatial, LIBERO-Object, LIBERO-Goal, LIBERO-90, and LIBERO-Long, each targeting distinct generalization capabilities such as spatial reasoning, object-level variation, goal specification, and long-horizon compositional manipulation. Following the standard evaluation protocol, we use the official LIBERO dataset and re-render all observations at a resolution of 256×256 to ensure a controlled comparison across methods. We train ODEWorld and all image-generation baselines on the full dataset.

AgiBot-World (Bu et al., 2025). For AgiBot-World, we train ODEWorld on the AgiBotWorldChallenge-2025 subset, a large-scale embodied manipulation benchmark built on the AgiBot World Colosseo platform (Bu et al., 2025). The subset used in our experiments contains approximately 30K manipulation trajectories, with an average length of around 260 frames per trajectory. It captures diverse robot-object interaction patterns and provides visual observations across a wide range of real-world manipulation tasks. Compared with existing short-horizon robot datasets, its relatively long trajectories and rich interaction dynamics make it well suited for continuous-time world modeling and future visual prediction.

LIBERO-LONG. As a subset of LIBERO, LIBERO-LONG focuses on long-horizon compositional manipulation, where each episode requires completing multiple sequential subgoals involving interactions with several objects. The benchmark consists of ten tasks: (1) Soup and sauce to basket, (2) Box and butter to basket, (3) Turn on the stove and place the pot, (4) Bowl to drawer, (5) Mugs on plates, (6) Book to rack, (7) Mug and pudding, (8) Soup and box to basket, (9) Both pots on the stove, and (10) Mug to microwave.

A.2 Real-World Settings

We follow recent SOTA practice (Black et al., 2024; Zheng et al., 2025; Feng et al.) to conduct real-world robotics experiments on a bi-manual AgileX robot equipped with two 6-DoF robotic arms, parallel-jaw grippers, and RGB cameras. We use X-VLA (Zheng et al., 2026b) as the policy backbone and incorporate ODEWorld-generated latent subgoals for sequential guidance. All experiments are conducted under the same training and evaluation protocol for a fair comparison with the original X-VLA policy. We evaluate ODEWorld on four manipulation tasks: (1) *Shrimp-to-pot*: Place a shrimp into a pot; (2) *Mouse packing*: Pack a mouse into a box; (3) *Pen insertion*: Grasp a thin pen and insert it into a narrow container, requiring fine-grained perception and precise control; (4) *Bi-manual rearrangement*: Simultaneously move objects using both arms in a complex scene, evaluating coordinated bi-manual control and multi-object dynamics modeling. Fig. 6 presents the initial states of these four real-world manipulation tasks.

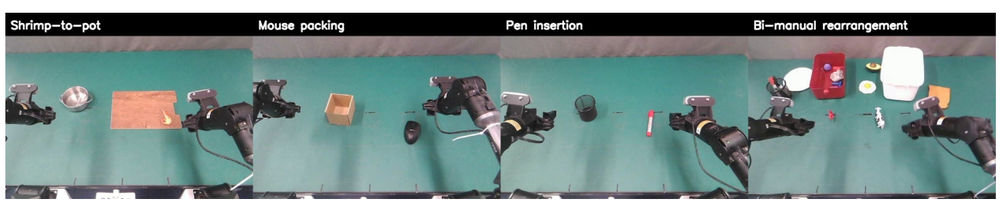


Figure 6 Initial states of the four real-world manipulation tasks. From left to right: *Shrimp-to-pot*, *Mouse packing*, *Pen insertion*, and *Bi-manual rearrangement*.

附录

A 实验细节

A.1 数据集

LIBERO（Liu et al., 2023）。LIBERO 是大型语言条件机器人操作基准，包含 130 项任务和 6,500 条高质量人类遥操作示范。任务覆盖抓取、抬升、放置、开启和多步物体重排等多样原子及组合技能。基准分为 LIBERO-Spatial、LIBERO-Object、LIBERO-Goal、LIBERO-90 和 LIBERO-Long 五个套件，分别面向空间推理、物体变化、目标指定和长时域组合操作等泛化能力。遵循标准评估协议，采用官方数据集，将所有观测重新渲染为 256×256 分辨率，以保证方法间受控比较。在完整数据集上训练 ODEWorld 和全部图像生成基线。

AgiBot-World（Bu et al., 2025）。在 AgiBot-World 中，用基于 AgiBot World Colosseo 平台（Bu et al., 2025）构建的大型具身操作基准 AgiBotWorldChallenge-2025 子集训练 ODEWorld。实验子集包含约 3 万条操作轨迹，每条平均约 260 帧，覆盖多样的机器人—物体交互模式，提供广泛真实操作任务的视觉观测。相较现有短时域机器人数据集，其较长轨迹与丰富交互动力学，非常适合连续时间世界建模及未来视觉预测。

LIBERO-LONG。作为 LIBERO 子集，LIBERO-LONG 聚焦长时域组合操作，每个回合需完成涉及多个物体交互的多个顺序子目标。基准含十项任务：（1）汤和酱入篮；（2）盒子和黄油入篮；（3）打开炉灶并放锅；（4）碗入抽屉；（5）杯子放盘子；（6）书放架子；（7）杯子和布丁；（8）汤和盒子入篮；（9）两个锅放炉灶；（10）杯子入微波炉。

A.2 真实世界设置

我们遵循近期最先进实践（Black et al., 2024; Zheng et al., 2025; Feng et al.），在配有两条 6 自由度机械臂、平行夹爪和 RGB 相机的 AgileX 双臂机器人上开展真实实验。采用 X-VLA（Zheng et al., 2026b）作为策略骨干，融入 ODEWorld 生成的潜在子目标进行序列引导。全部实验遵循相同训练与评估协议，以公平比较原始 X-VLA 策略。评估四项操作任务：（1）虾入锅：将虾放入锅；（2）鼠标装盒：将鼠标装入盒子；（3）插笔：抓取细笔并插入狭窄容器，要求细粒度感知和精确控制；（4）双臂重排：在复杂场景中同时用双臂移动物体，评估协调的双臂控制与多物体动力学建模。图 6 展示四项真实操作任务的初始状态。

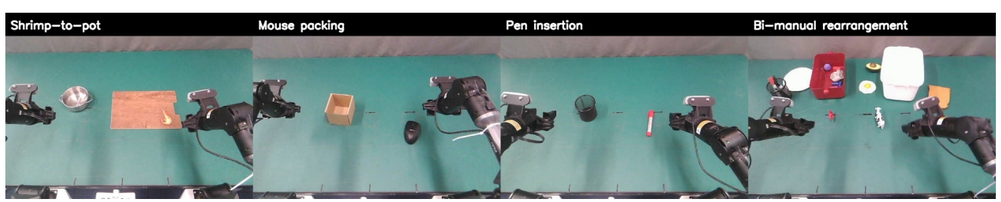


图 6：四项真实机器人操作任务的初始状态。从左至右：虾入锅、鼠标装盒、插笔和双臂重排。

A.3 Video Generation Baselines

V-JEPA 2 (Assran et al., 2025). We adopt the pre-trained V-JEPA 2 ViT-L/16 checkpoint as the visual representation backbone and fine-tune it on LIBERO demonstration videos from the agentview camera. The model follows the standard online-target encoder design of V-JEPA 2, where the target encoder is updated by EMA. During training, we use short video clips cropped to 256×256 and apply a first-tubelet-visible masking strategy, forcing the predictor to infer future latent tokens from the initial visible spatio-temporal context. To incorporate task information, we extract language features with a frozen CLIP ViT-B/32 text encoder (Radford et al., 2021) and inject them into the predictor through zero-initialized cross-attention layers. Since V-JEPA 2 predicts latent representations rather than pixels, we additionally train a lightweight convolutional decoder for RGB reconstruction. The encoder-predictor stack is frozen when training this decoder. To improve robustness to errors in predicted latents, we perturb the training latents with Gaussian noise. The decoder is optimized with an L1 reconstruction loss over video frames and an LPIPS (Zhang et al., 2018b) loss on temporally subsampled frames.

LDP (Xie et al., 2025). We implement LDP based on the official codebase and make several modifications for fair comparison under high-resolution LIBERO video modeling. Specifically, we increase the visual input resolution to 256×256 and enlarge the visual token budget in the planner/IDM pipeline accordingly. To handle the high-dimensional SD-VAE latent representation, we introduce a spatial convolutional projection before policy learning, which compresses the latent feature map and stabilizes training. We also add explicit goal-image conditioning by encoding the goal frame with the same VAE and injecting the goal latent into the planning module. During inference, LDP performs open-loop rollout, where future latent trajectories are autoregressively predicted without intermediate environment feedback. These settings are used consistently for all reported LDP results.

A.4 Policy Learning Baselines

GLCBC. GLCBC is a goal-conditioned language behavioral cloning baseline implemented within the LBP policy architecture but without the high-level planner. The current image observation is encoded by a ResNet-34 (He et al., 2016), while the language instruction is encoded by CLIP (Radford et al., 2021). In addition, we use a predefined final goal image as goal conditioning and encode it with the DecisionNCE image encoder (Li et al., 2024b). The language and goal-image embeddings are concatenated and injected into the policy through a FiLM (Perez et al., 2018) module to modulate the visual features. Finally, the low-level diffusion policy takes the conditioned visual semantics and current proprioception as inputs, and predicts the diffusion noise for action generation.

SuSIE (Black et al., 2024). We utilize the official codebase with minimal modifications, altering only the datasets. For simulation experiments, we fine-tune the released SuSIE checkpoint on the LIBERO dataset.

Seer (Tian et al., 2025). We use Seer as a vision-language-action baseline following its original implementation. The model predicts future visual representations from the current observation and language instruction, and then uses an inverse-dynamics head to predict actions conditioned on both the current observation and the predicted future features.

VPP (Hu et al., 2025). VPP learns an implicit inverse dynamics model conditioned on predicted future representations inside VDMs. To predict a more precise future, VPP fine-tunes the pre-trained video foundation model on robot datasets along with internet human manipulation data. We follow the official codebase to finetune it on the LIBERO dataset.

A.5 Computation Resource

We train models on 8 A100 GPUs and evaluate infernece time in Tab. 1 on 1 A100 GPU.

A.3 视频生成基线

V-JEPA 2（Assran et al., 2025）。我们采用预训练的 V-JEPA 2 ViT-L/16 检查点作为视觉表征骨干，并在 LIBERO 的 agentview 相机示范视频上微调。模型遵循 V-JEPA 2 标准在线 — 目标编码器设计，目标编码器通过指数移动平均更新。训练中使用裁剪至 256×256 的短视频片段，并采用仅首个管状块可见的掩码策略，迫使预测器根据初始可见时空上下文推断未来潜在标记。为引入任务信息，用冻结的 CLIP ViT-B/32 文本编码器（Radford et al., 2021）提取语言特征，并通过零初始化的交叉注意力层注入预测器。由于 V-JEPA 2 预测潜在表征而非像素，我们额外训练轻量卷积解码器重建 RGB；训练该解码器时冻结编码器—预测器系统。为增强对预测潜变量误差的鲁棒性，给训练潜变量加入高斯噪声。解码器使用视频帧上的 L1 重建损失，以及时间子采样帧上的 LPIPS（Zhang et al., 2018b）损失优化。

LDP（Xie et al., 2025）。基于官方代码实现 LDP，并作若干修改，以在高分辨率 LIBERO 视频建模下公平比较。将视觉输入分辨率提高至 256×256，相应扩大规划器／逆动力学模型流水线的视觉标记预算。为处理高维 SD-VAE 潜在表征，在策略学习前引入空间卷积投影，压缩潜在特征图并稳定训练。还用同一 VAE 编码目标帧，将目标潜变量注入规划模块，以加入显式目标图像条件。推断时，LDP 进行开环生成，自回归预测未来潜在轨迹，期间不接收环境反馈。所有报告的 LDP 结果均一致采用这些设置。

A.4 策略学习基线

GLCBC。GLCBC 是在 LBP 策略架构内实现、但不含高层规划器的目标与语言条件行为克隆基线。当前图像观测由 ResNet-34（He et al., 2016）编码，语言指令由 CLIP（Radford et al., 2021）编码。此外，以预定义的最终目标图像作为目标条件，用 DecisionNCE 图像编码器（Li et al., 2024b）编码。语言与目标图像嵌入拼接后，通过 FiLM（Perez et al., 2018）模块注入策略，调制视觉特征。最后，低层扩散策略接收条件化视觉语义和当前本体感知，预测用于动作生成的扩散噪声。

SuSIE（Black et al., 2024）。使用官方代码，仅作最小修改，更换数据集。模拟实验中，在 LIBERO 数据集上微调公开的 SuSIE 检查点。

Seer（Tian et al., 2025）。遵循原始实现，采用 Seer 作为视觉—语言—动作基线。模型从当前观测和语言指令预测未来视觉表征，再用逆动力学头，以当前观测及预测未来特征为条件预测动作。

VPP（Hu et al., 2025）。VPP 在视频扩散模型内部，以预测未来表征为条件学习隐式逆动力学模型。为更精确预测未来，利用机器人数据和互联网人类操作数据微调预训练视频基础模型。我们遵循官方代码在 LIBERO 数据集上微调。

A.5 计算资源

在 8 张 A100 GPU 上训练模型，表 1 的推断时间在 1 张 A100 GPU 上评估。

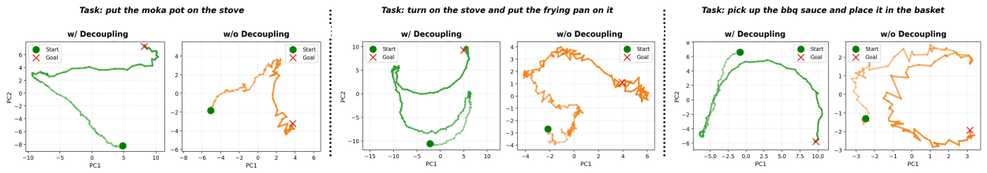


Figure 7 Latent trajectory visualization of ablation on dynamical representation decoupling.

Table 4 Ablations on dynamical representation decoupling and first-order supervision

Method	Short Fragment @16frames			Long Fragment @64frames		
	PSNR↑	LPIPS↓	FPS↑	PSNR↑	LPIPS↓	FPS↑
w/o dynamical representation decoupling	**20.65**	0.127	7.301	**19.53**	0.212	2.290
w/o first-order supervision	12.71	0.751	**33.67**	-	-	-
ODEWorld (Ours)	20.53	**0.109**	**33.67**	19.46	**0.134**	**13.83**

B Ablation Studies

B.1 Ablation on Dynamical Representation Decoupling

To validate the effectiveness of our architectural design, we evaluate the impact of the dynamical representation decoupling module in ODEWorld. We observe that without explicitly isolating the static context s_0 from the state space, the learned latent trajectories become significantly non-smooth and jittery as shown in Fig. 7. The primary reason is that not all nuanced visual features in a raw video observation can be seamlessly modeled as an ODE function. Raw feature spaces inherently contain a substantial amount of noisy, time-invariant, and non-ODE-compatible information, such as static backgrounds and object textures. Forcing a continuous velocity field to fit these static details fundamentally disrupts the dynamics learning process. As shown in Tab. 4, the proposed decoupling strategy not only significantly improves inference efficiency (significantly higher FPS) but also enhances perceptual consistency (much lower LPIPS scores, especially for long-horizon prediction). We note that the non-decoupled variant attains marginally higher PSNR, which is expected rather than contradictory: as an MSE-based pixel-level metric, PSNR is dominated by large static regions (e.g., backgrounds and textures) that this variant retains and reconstructs faithfully, whereas its worse LPIPS reflects the temporal and semantic inconsistency that such per-frame fidelity does not capture. Consequently, without decoupling, the latent structure fails to effectively scaffold the direct first-order supervision to optimize the velocity field. By introducing the initial state-conditioned encoder and decoder, our decoupling mechanism ensures that the latent representation focuses exclusively on essential dynamical changes, which is crucial for learning a well-behaved, temporally smooth, and physically meaningful latent flow.

B.2 Ablation on First-Order Supervision

To verify the effectiveness of the proposed first-order supervision, we implement consistency loss below to replace the reconstruction loss $\mathcal{L}_{\text{dyn-recon}}$ and velocity loss $\mathcal{L}_v$:

$$\mathcal{L}_{\text{consistency}} = \mathbb{E}_{s_0,s_t\sim\mathcal{D},k\in\{1,\cdots,K\},t}\left\| g_{\text{dyn}}\left(z_t + \int_t^{t+k} v_\theta(z_t,t;z_0,c)dt; s_0\right) - s_{t+k}\right\|^2 \tag{7}$$

where $z_t = f_{\text{dyn}}(s_t; s_0)$, $z_0 = f_{\text{dyn}}(s_0; s_0)$ and $K = 4$. Specifically, this consistency-loss variant minimizes the error of reconstructing the subsequent K frames obtained by planning through the ODE latent space, which has been adopted in many works on neural ODE (Chen et al., 2018; Rubanova et al., 2019). In Tab. 4, we observe that it suffers from optimization difficulty, yielding poor short-horizon prediction and visually implausible long-horizon rollouts (so we fill those results in Tab. 4 with "-"). We attribute this to

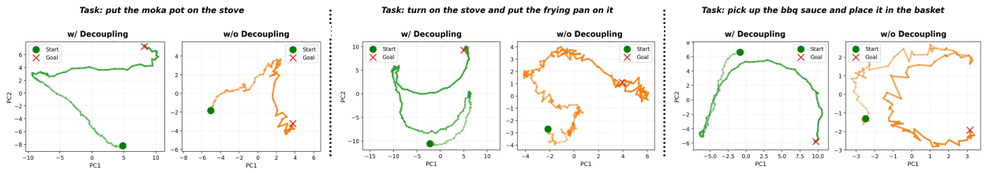


图 7：动力学表征解耦消融实验的潜在轨迹可视化。

表 4：动力学表征解耦和一阶监督的消融。

方法	短片段：16 帧			长片段：64 帧		
	PSNR↑	LPIPS↓	帧秒↑	PSNR↑	LPIPS↓	帧秒↑
无动力学表征解耦	**20.65**	0.127	7.301	**19.53**	0.212	2.290
无一阶监督	12.71	0.751	**33.67**	-	-	-
ODEWorld（本文）	20.53	**0.109**	**33.67**	19.46	**0.134**	**13.83**

B 消融研究

B.1 动力学表征解耦消融

为验证架构设计的有效性，我们评估 ODEWorld 中动力学表征解耦模块的影响。若不从状态空间显式分离静态上下文 s_0，学得的潜在轨迹会明显不平滑并发生抖动，如图 7 所示。主要原因在于，原始视频观测中的细微视觉特征，并非都能自然建模为 ODE 函数。原始特征空间固有地包含大量噪声、不随时间变化且不兼容 ODE 的信息，如静态背景和物体纹理。强迫连续速度场拟合这些静态细节，会从根本上干扰动力学学习。表 4 显示，解耦策略既显著提高推断效率（FPS 明显更高），也增强感知一致性（LPIPS 显著更低，尤其在长时域预测时）。未解耦变体的 PSNR 略高，这符合预期而非矛盾：PSNR 是基于均方误差的像素级指标，主要受背景和纹理等大面积静态区域支配，未解耦变体保留并忠实重建了它们；而较差的 LPIPS 反映出逐帧保真度无法捕捉的时间与语义不一致。因此，不解耦时，潜在结构无法有效支撑用于优化速度场的直接一阶监督。引入以初始状态为条件的编码器和解码器后，解耦机制确保潜在表征只关注关键动力学变化，这对学习行为良好、时间平滑且具有物理意义的潜在流至关重要。

B.2 一阶监督消融

为验证一阶监督的有效性，实现以下一致性损失，替换重建损失 $\mathcal{L}_{\text{dyn-recon}}$ 与速度损失 $\mathcal{L}_v$:

$$\mathcal{L}_{\text{consistency}} = \mathbb{E}_{s_0,s_t\sim\mathcal{D},k\in\{1,\cdots,K\},t}\left\| g_{\text{dyn}}\left(z_t + \int_t^{t+k} v_\theta(z_t,t;z_0,c)dt; s_0\right) - s_{t+k}\right\|^2 \tag{7}$$

其中 $z_t = f_{\text{dyn}}(s_t;\ s_0)$、$z_0 = f_{\text{dyn}}(s_0;\ s_0)$ 及 $K = 4$。具体而言，该一致性损失变体最小化经 ODE 潜空间规划得到的后续 K 帧的重建误差；许多神经 ODE 工作采用这种方式（Chen et al., 2018; Rubanova et al., 2019）。表 4 中观察到它存在优化困难，短时域预测较差，长时域生成在视觉上不合理（因此表 4 相关结果填为"–"）。我们认为原因在于

Table 5 Comparisons on the choices of vision feature encoder.

Encoder	Short Fragment @16frames		Long Fragment @64frames	
	PSNR↑	LPIPS↓	PSNR↑	LPIPS↓
LIV	19.90	0.158	18.25	0.237
SigLIP 2	20.38	0.133	19.21	0.168
DINOv2 (Ours)	**20.53**	**0.109**	**19.46**	**0.134**

Table 6 Ablation on the number of latent tokens.

# Tokens	Short Fragment @16frames		Long Fragment @64frames	
	PSNR↑	LPIPS↓	PSNR↑	LPIPS↓
4	**20.99**	0.114	**19.68**	0.148
1 (Ours)	20.53	**0.109**	19.46	**0.134**

two aspects. First, the consistency loss couples two distinct objectives — prediction and reconstruction — forcing the model to learn a latent space that is simultaneously reconstructable and ODE-plannable. Second, this coupling is especially harmful in our setting: unlike standard dynamical systems where the goal is to fit one specific governing ODE, ODEWorld only needs to fit some admissible ODE in the latent space, so the target is underdetermined and the entangled objective adds too strict constraints to settle on a consistent solution. In contrast, ODEWorld decouples these objectives via a reconstruction loss and a direct first-order supervision loss, optimizing each independently, which leads to more stable optimization and, in turn, better video generation quality.

B.3 Ablations on Vision Encoders

We examine how ODEWorld is compatible with different choices of vision feature extractors by replacing DINOv2 with SigLIP 2 (Tschannen et al., 2025) and LIV (Ma et al., 2023). As shown in Tab. 5, ODEWorld maintains robust video generation quality across all three encoders. While the gaps are small, DINOv2 achieves the best overall performance on both short- and long-horizon prediction, consistent with its widely recognized effectiveness as a visual encoder.

B.4 Ablation on Token Numbers

We increase the number of tokens from 1 to 4 and keep other configurations unchanged. Results in Tab. 6 show comparable performance between 1- and 4-token models, indicating that a *single* latent token is already sufficient to achieve strong performance across the evaluated scenarios. This also indicates that our key designs make ODEWorld capable of efficiently capturing the underlying dynamics from pixels with very compact latent representation.

B.5 Ablation on Goal Descriptions

ODEWorld naturally supports image goals, where the target image provides explicit visual guidance for future dynamics prediction. However, in practical scenarios, goals are often specified by language instructions. To extend ODEWorld to language-based goal descriptions, we introduce a lightweight goal image predictor to bridge the language and visual modalities.

Specifically, since ODEWorld adopts DINOv2 as the visual backbone rather than a large-scale vision-language model, we avoid introducing additional multimodal alignment complexity into the dynamics model. Instead, we train a lightweight goal image predictor consisting of four cross-attention layers, which takes the current observation and language instruction as inputs and predicts the corresponding goal image. As shown in Fig. 8, the goal image predictor generates semantically meaningful goal images from the current observation and language instruction, providing effective visual guidance for ODEWorld.

表 5：视觉特征编码器选择的比较。

编码器	短片段：16 帧		长片段：64 帧	
	PSNR↑	LPIPS↓	PSNR↑	LPIPS↓
LIV	19.90	0.158	18.25	0.237
SigLIP 2	20.38	0.133	19.21	0.168
DINOv2（本文）	**20.53**	**0.109**	**19.46**	**0.134**

表 6：潜在标记数量的消融。

标记数	短片段：16 帧		长片段：64 帧	
	PSNR↑	LPIPS↓	PSNR↑	LPIPS↓
4	**20.99**	0.114	**19.68**	0.148
1（本文）	20.53	**0.109**	19.46	**0.134**

两个方面。第一，一致性损失耦合预测与重建这两个不同目标，迫使模型学习同时可重建、可用 ODE 规划的潜空间。第二，这种耦合在本设置中特别有害：标准动力系统旨在拟合某个特定支配 ODE，而 ODEWorld 只需在潜空间拟合某个可接受的 ODE，因此目标欠确定，纠缠的目标施加过严约束，难以收敛到一致解。ODEWorld 则用重建损失和直接一阶监督损失解耦二者，分别优化，从而优化更稳定，视频生成质量也更好。

B.3 视觉编码器消融

通过把 DINOv2 替换为 SigLIP 2（Tschannen et al., 2025）和 LIV（Ma et al., 2023），检验 ODEWorld 与不同视觉特征提取器的兼容性。表 5 表明，在三个编码器上均保持稳健的视频生成质量。差距虽小，DINOv2 的短、长时域预测整体最好，与其公认的视觉编码有效性一致。

B.4 标记数量消融

将标记数量从 1 增至 4，其他配置不变。表 6 中，两种模型表现相当，表明单个潜在标记已足以在评估场景中取得较强性能。这也说明，关键设计使 ODEWorld 能用极紧凑潜在表征，高效捕捉像素中的底层动力学。

B.5 目标描述消融

ODEWorld 天然支持图像目标，由目标图像为未来动力学预测提供显式视觉引导。但实际场景中，目标通常由语言指令指定。为支持语言目标描述，引入轻量目标图像预测器，连接语言和视觉模态。

由于 ODEWorld 的视觉骨干是 DINOv2 而非大型视觉语言模型，我们避免给动力学模型引入额外多模态对齐复杂度。转而训练由四层交叉注意力组成的轻量目标图像预测器，接收当前观测和语言指令，预测对应目标图像。图 8 显示，它能生成有语义意义的目标图像，为 ODEWorld 提供有效视觉引导。

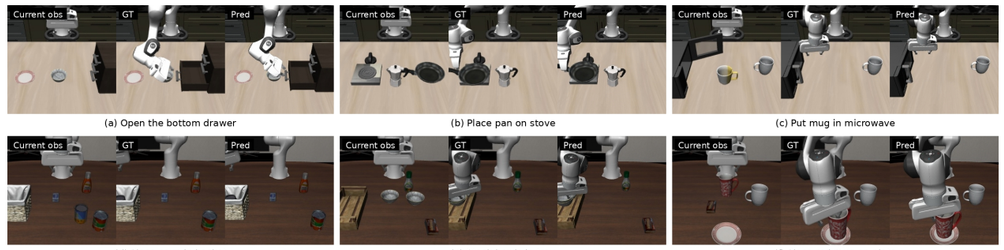


Figure 8 Qualitative results of the goal image predictor. Given the current observation and a language instruction, the goal image predictor generates a semantically consistent goal image for ODEWorld.

Table 7 Comparison between ground-truth image goals and predicted language goals.

Goal Conditioning	Short Fragment @16frames		Long Fragment @64frames	
	PSNR↑	LPIPS↓	PSNR↑	LPIPS↓
Predicted Goal Image	21.06	0.110	19.91	0.142
GT Goal Image	20.53	0.109	19.46	0.134

Table 8 Ablation on the detach in velocity learning.

Methods	Short Fragment @16frames		Long Fragment @64frames	
	PSNR↑	LPIPS↓	PSNR↑	LPIPS↓
w/o detach	21.18	0.107	20.00	0.123
w/ detach	20.53	0.109	19.46	0.134

We further evaluate ODEWorld using the predicted goal images for goal conditioning and compare it with the standard setting using ground-truth goal images. As reported in Tab. 7, the two settings achieve nearly identical video prediction performance on both short- and long-horizon rollouts. These results demonstrate that the proposed goal image predictor effectively bridges language and visual goal representations, enabling ODEWorld to naturally support language-specified goals without sacrificing prediction quality.

B.6 ODEWorld without Detach in Velocity Learning

In PT-Flow, we use the stop-gradient operation $\mathtt{sg}(\cdot)$ on the target latent velocity to stabilize the optimization, following common practices in representation learning (Chen and He, 2021). Interestingly, we find that PT-Flow remains effective even without stop-gradient, as shown in Tab. 8. This suggests that the proposed objective itself can provide effective constraints for learning meaningful representations, as the first-order velocity constraint directly regularizes the temporal evolution in latent space.

C Analysis of Representation Collapse

Following RankMe (Garrido et al., 2023), we use the effective rank of mean-centered representations to assess potential representation collapse. We randomly sample 5,000 representations across planning horizons $h \in \{1, 4, 8, 16, 32\}$. As shown in Tab. 9, ODEWorld's 768-dimensional dynamics latent z consistently maintains an effective rank above 400 across all horizons. In Tab. 10, we further compare ODEWorld with V-JEPA 2 representations and DINOv2 CLS features under the same sampling protocol. ODEWorld achieves a substantially higher effective rank than V-JEPA 2 and DINOv2, indicating that its learned latent dynamics representations remain diverse and expressive with a lower risk of representation collapse.

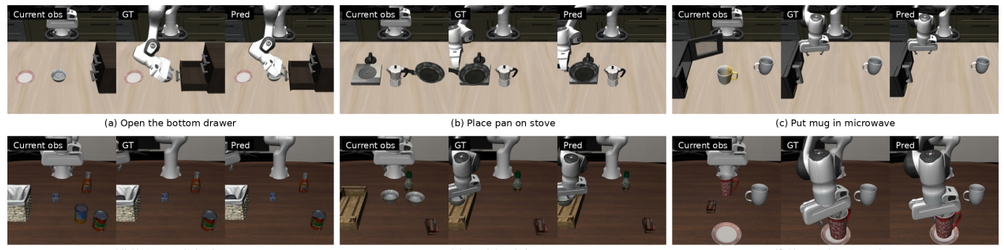


图 8：目标图像预测器的定性结果。给定当前观测和语言指令，预测器为 ODEWorld 生成语义一致的目标图像。

表 7：真实图像目标与预测语言目标的比较。

目标条件	短片段：16 帧		长片段：64 帧	
	PSNR↑	LPIPS↓	PSNR↑	LPIPS↓
预测目标图像	21.06	0.110	19.91	0.142
真实目标图像	20.53	0.109	19.46	0.134

表 8：速度学习中 detach 的消融。

方法	短片段：16 帧		长片段：64 帧	
	PSNR↑	LPIPS↓	PSNR↑	LPIPS↓
不使用 detach	21.18	0.107	20.00	0.123
使用 detach	20.53	0.109	19.46	0.134

进一步以预测目标图像作为目标条件，比较使用真实目标图像的标准设置。表 7 中，两种设置在短、长时域生成上的视频预测表现几乎相同。这表明目标图像预测器有效连接语言与视觉目标表征，使 ODEWorld 自然支持语言指定目标而不牺牲预测质量。

B.6 速度学习不使用 detach 的 ODEWorld

遵循表征学习常见实践（Chen and He, 2021），PT-Flow 对目标潜在速度采用停止梯度操作 sg（·），以稳定优化。有趣的是，表 8 显示，不停止梯度也依然有效。这说明目标自身提供了学习有意义表征的有效约束，因为一阶速度约束直接正则化潜空间中的时间演化。

C 表征坍塌分析

遵循 RankMe（Garrido et al., 2023），以减去均值后的表征有效秩评估潜在坍塌。在规划时域 $h \in \{1, 4, 8, 16, 32\}$上随机采样 5,000 个表征。表 9 显示，ODEWorld 的 768 维动力学潜变量 z在所有时域始终保持 400 以上的有效秩。表 10 进一步在相同采样协议下比较 ODEWorld、V-JEPA 2 表征及 DINOv2 CLS 特征。ODEWorld 的有效秩显著更高，表明学得的潜在动力学表征保持多样性和表达能力，表征坍塌风险更低。

Table 9 Centered RankMe effective rank analysis of ODEWorld dynamics latents across different planning horizons.

Horizon h	1	4	8	16	32
Effective Rank / Dim.	417.2/768	413.9/768	422.1/768	433.1/768	439.5/768

Table 10 Centered RankMe effective rank comparison with V-JEPA 2 and DINOv2 representations.

Representation	Dimension	Effective Rank (Mean)
V-JEPA 2	1024	203.7
DINOv2 CLS	768	376.1
ODEWorld z	768	425.2

D Additional Results

D.1 Additional Video Generation Quality Comparisons

In Fig. 9, we provide additional comparisons for long-horizon video generation in the LIBERO simulation environment. These visualizations complement the main paper and further illustrate that ODEWorld can produce temporally coherent and physically plausible rollouts, while baseline methods often suffer from visual degradation and temporal inconsistency over extended horizons.

D.2 Additional Bidirectional Video Generation Results

In Fig. 10, we show additional bidirectional video generation results on both LIBERO and AgiBot World. We perform closed-loop planning that updates the initial state every integration step $\tau = 0.05$. When we plan backward, we integrate velocity $-v_\theta(z_0, 0; z_0, c)$ for every integration step and reset the planned latent state as z_0 for the next integration step.

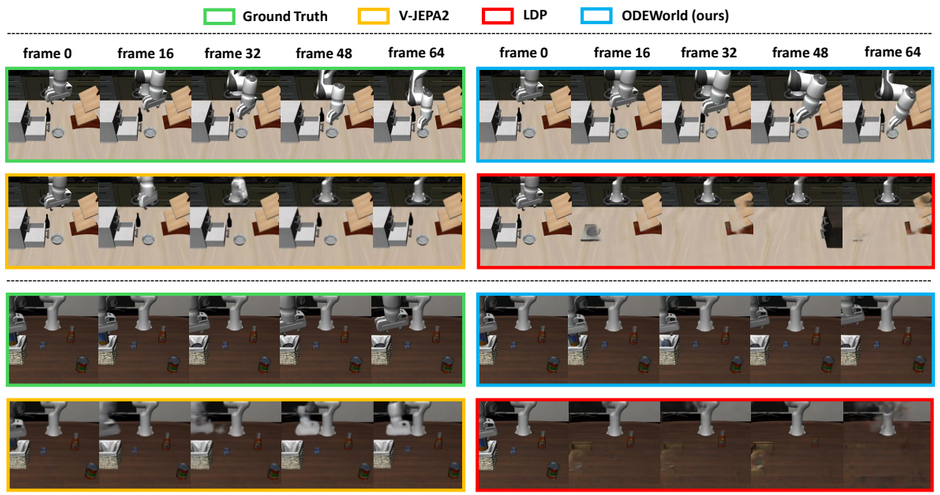


Figure 9 ODEWorld enables long-horizon video generation in the LIBERO simulation environment, producing temporally coherent and physically plausible videos that closely match the ground-truth observations. In contrast, our baseline methods LDP and V-JEPA 2 often suffer from blurry visual predictions and dataset-induced hallucinations in long-horizon generation, especially at later frames, such as frame 64.

表 9：中心化 RankMe 不同规划时域下 ODEWorld 动力学潜变量的有效秩分析

时域 h	1	4	8	16	32
有效秩／维度	417.2/768	413.9/768	422.1/768	433.1/768	439.5/768

表 10：与 V-JEPA 2 和 DINOv2 表征比较经中心化的 RankMe 有效秩。

表征	维度	有效秩（均值）
V-JEPA 2	1024	203.7
DINOv2 CLS	768	376.1
ODEWorld z	768	425.2

D 更多结果

D.1 更多视频生成质量比较

图 9 提供 LIBERO 模拟环境中更多长时域视频生成比较。这些可视化补充正文，进一步说明 ODEWorld 能产生时间连贯、物理可信的生成轨迹，而基线在长时域中常出现视觉退化与时间不一致。

D.2 更多双向视频生成结果

图 10 给出 LIBERO 和 AgiBot World 上更多双向视频生成结果。采用闭环规划，每个积分步 $\tau = 0.05$更新初始状态。反向规划时，每步对速度 $-v_\theta$（z_0、0； z_0、 c）积分，并将规划潜在状态重置为 z_0，用于下一积分步。

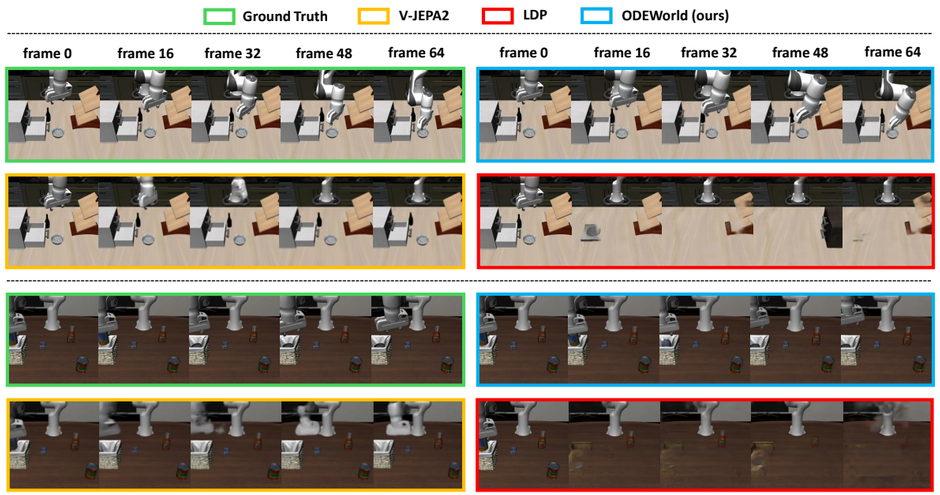


图 9：ODEWorld 在 LIBERO 模拟环境中实现长时域视频生成，视频时间连贯、物理可信，并紧贴真实观测。相比之下，LDP 和 V-JEPA 2 基线常在长时域生成中出现模糊视觉预测及数据集诱发的幻觉，尤其是第 64 帧等后期帧。

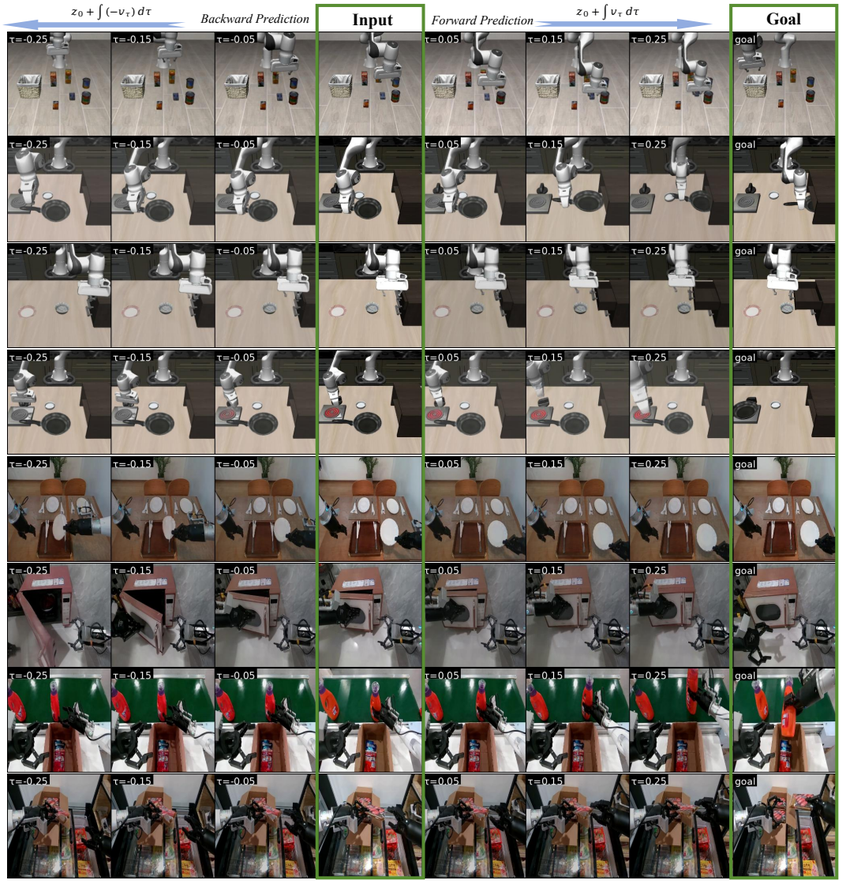


Figure 10 Additional visualization results of bidirectional prediction.

E Limitations

Although our proposed framework provides a promising architecture for continuous-time predictive modeling, the current version of ODEWorld still has several limitations and can be further enhanced in future research. First, while the proposed ODEWorld demonstrates some impressive capabilities and efficiency, its scale and amount of training data (two robotics datasets) remain small as compared to many large video-based world models. This limitation stems primarily from computational resource constraints. Future studies can be conducted to further scale the model size and train on larger and more diverse datasets. Second, our current version of ODEWorld uses a frozen DINOv2 (Oquab et al., 2024) image encoder to encode raw observations. As DINO is designed to encode single-frame images without much consideration for temporal consistency, it could produce noisy and temporally inconsistent embeddings for sequential modeling. In our study, we introduce the Savitzky–Golay derivative filter (Saramäki et al., 1993) to enhance the smoothness and approximation quality of the ground truth velocity target to mitigate the deficiencies of the DINO encoder. Further investigations can consider replacing DINO with an image encoder with temporal consistency design (Li et al., 2024b) or jointly tuning the DINO encoder during model training. Lastly, the current version of ODEWorld does not incorporate action conditioning and primarily focuses on video generation. Future studies can consider injecting action conditioning into a pre-trained ODEWorld model, which can further unlock its capability for control tasks.

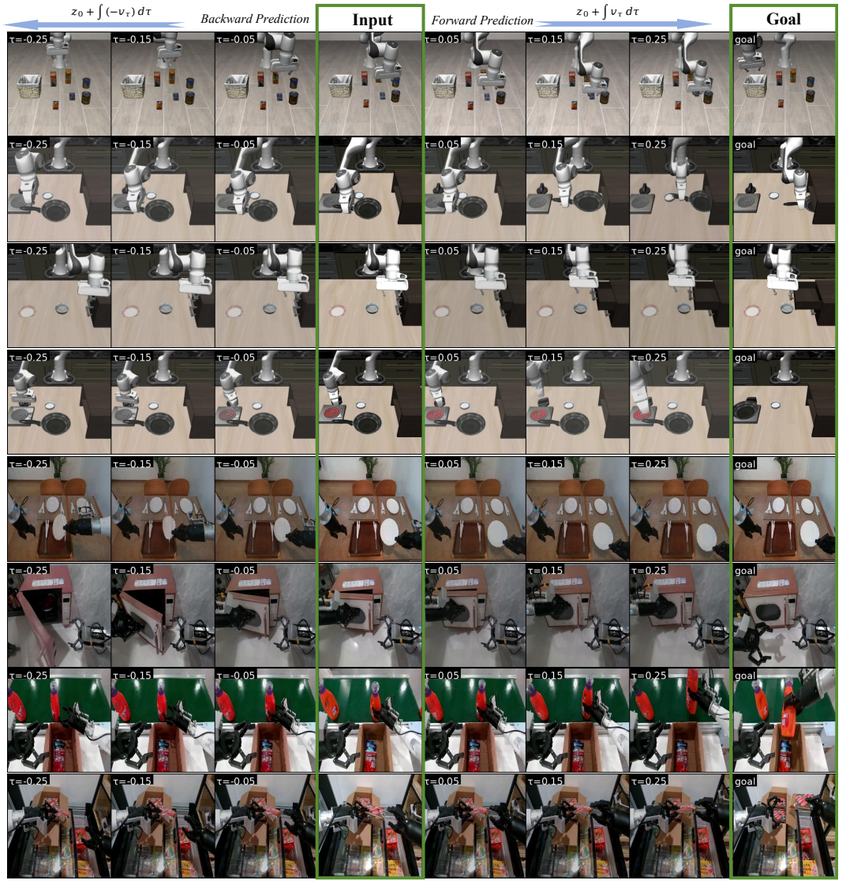


图 10：更多双向预测可视化结果。

E 局限性

尽管我们的框架为连续时间预测建模提供了有前景的架构，当前 ODEWorld 仍有若干局限，后续研究可进一步改进。首先，虽然模型表现出令人印象深刻的能力与效率，其规模及训练数据量（两个机器人数据集）仍远小于许多大型视频世界模型。这主要来自计算资源限制，未来可扩大模型并在更大、更多样的数据集上训练。其次，当前版本使用冻结的 DINOv2（Oquab et al., 2024）图像编码器编码原始观测。DINO 面向单帧图像，较少考虑时间一致性，因此可能产生带噪声、时间不一致的序列嵌入。我们引入 Savitzky–Golay 导数滤波器（Saramäki et al., 1993），提升真实速度目标的平滑性与近似质量，缓解 DINO 编码器的不足。后续可考虑换用具有时间一致性设计的图像编码器（Li et al., 2024b），或在训练时联合调优 DINO。最后，当前版本未加入动作条件，主要聚焦视频生成。未来可在预训练 ODEWorld 中注入动作条件，进一步解锁控制任务能力。